

Cute Goldbach Gaps

Extremal Least Goldbach Summands:
Constructions, Costs, and Closure

A thesis on the engineering of Goldbach deserts

Anonymous

July 2026

Every integer, congruence system, primality certificate, verifier, and search log referenced in this thesis is committed to the accompanying artifact repository and can be re-checked from scratch on a laptop. Nothing rests on a log we could not regenerate.

Abstract

For an even integer N , the *least Goldbach summand* is $\mathbf{g}(N) = \min\{p \text{ prime} : N - p \text{ prime}\}$ — the quantity $p(n)$ of the minimal Goldbach partition studied by Granville, van de Lune and te Riele. Nature keeps it tiny: exhaustive verification gives $\mathbf{g}(N) \leq 9\,781$ for every even $N \leq 4 \cdot 10^{18}$. This thesis is about the opposite, extremal and constructive question: *how large can $\mathbf{g}(N)$ be forced to be, and at what cost in the size of N ?*

Three record games organise the answer. The *height game* maximises $\mathbf{g}(N)$ with no size limit; the *threshold game* minimises N subject to $\mathbf{g}(N) > Q$; the *budget game* maximises $\mathbf{g}(N)$ subject to $N < X$. After proving unconditionally that $\mathbf{g}$ is unbounded and transferring the Ford–Green–Konyagin–Maynard–Tao long-gap machinery into an explicit lower bound $\mathbf{g}(N) \gg \log N \log_2 N \log_4 N / \log_3 N$ along even integers with exhibited partitions, the thesis develops the partial-cover search paradigm — congruence covers of the prime offsets, Chinese-remainder progressions, and probable-prime scanning — into a measured, predictive science with a validated cost model, and uses it to set every record it then studies. The certified board: $\mathbf{g}(N) = 1\,157\,341$ at 2480 digits for the height game, exceeding the longest prime gap ever certified while using integers seven times shorter than that gap’s own endpoints; a 150-digit N with $\mathbf{g}(N) = 104\,527$ for the threshold game $T(100\,000)$, forty-nine digits below the first-generation record; and $\mathbf{g}(N) = 119\,419$ below 10^{199} for the budget game. A companion construction of 2692 digits carries a Goldbach desert 126.5 times longer than its fully-certified local prime gap, separating the two phenomena experimentally.

The second half of the thesis asks why the records stop where they do, and answers with mathematics rather than fatigue. The search exponent is governed by a density model validated to ± 0.1 nats across $1.2 \cdot 10^9$ scanned candidates; the digit frontier at fixed threshold is an explicit, measured curve; and beneath both lie provable barriers. Compressing constructions below ~ 100 digits reduces to modular subset-sum with per-position choices at density ≈ 1.026 — the regime where lattice reduction and Wagner-style list merging both fail — and to Chinese-remainder list decoding beyond the Johnson radius, which practical lattice reduction does not penetrate even on solution-abundant instances (measured). A closure theorem for compositeness-certificate schemes then shows, across a concrete grammar of algebraic mechanisms swept by 13661 hand-enumerated and 41 LLM-evolved instances, that congruence covering is the unique economical mechanism: every alternative family in the grammar is closed by a proof of its own kind, and the game’s whole exponent is an integrality gap that survives exact optimization over half the cover at once. What remains open is named precisely: a rounding-gap programme importing the Erdős–Rankin covering technology constructively, a sub-modexp compositeness-detection question posed to complexity theory, two coding-theoretic problems, and a two-sided construction that would uncap the height game by making the partition’s primality certificate free. The thesis closes with the walls each game will hit under any plausible compute, and the records’ honest epitaph: upper bounds, manufactured, falling toward a floor they will never touch.

Preface: how to read this thesis

Each chapter is written to stand alone as a paper, with its own abstract, and the chapters are ordered as the research was: construction first, measurement second, impossibility last.

- **Chapter 1** defines the least Goldbach summand, the empirical background, and the three record games, and states the final record board.
- **Chapter 2** is the foundations paper: factorial and Wilson obstructions, primordial deserts, unconditional unboundedness, prime-offset congruence covers, and the transfer of the large-prime-gap lower bounds. It subsumes, updates, and supersedes the July 2026 manuscript *Cute Goldbach Gaps*, whose first-generation records ($\mathfrak{g} = 109\,621$ at 237 digits, $\mathfrak{g} = 105\,667$ at 199 digits) are the baseline everything later improves.
- **Chapter 3** builds the quantitative theory of partial-cover search: the density model, its five-scale validation, the digit-compute exchange rate, the frontier curves, and the statistical discipline that kept negative results honest.
- **Chapter 4** is the records paper: two independent pipelines, the search engine and its measured optimizations, the certification standard, and the certified ladder from 199 digits down to 150.
- **Chapter 5** compares Goldbach deserts with ordinary prime gaps and reports the height-game records, including the desert that is 126.5 times longer than its own certified prime gap.
- **Chapter 6** proves the compression barriers: why the threshold record cannot be pushed below ~ 100 digits by any known technique, with the failed attacks measured rather than presumed.
- **Chapter 7** states and proves (at draft rigor, on a concrete grammar) the Divisor-Paradigm Closure Theorem: within a language of planted compositeness certificates, nothing beats covering — and reports the adversarial falsification engine that tried to break it, including the accounting hole it found in our own referee first.
- **Chapter 8** collects what is proved, what is measured, what is conjectured, and the open problems, ranked.
- **Appendix A** maps every claim to its machine-checkable artifact and gives the verification runbook.

Three writing conventions, kept throughout. First, notation is explained in words at first use — the intended reader is a mathematically literate non-specialist, and no step should require outside references to follow. Second, every record claim separates three obligations: (*a*) no small prime

is a summand, *(b)* a Goldbach partition exists, *(c)* the least summand equals the stated value; witnesses for (a) are unconditional, and (b) always means an exhibited partition with a certified prime complement, never an appeal to a conjecture. Third, negative results — campaigns that found nothing, attacks that failed, one retracted claim — are reported with the same care as the records, because the thesis’s second half is built out of them.

The research was carried out in 2026 across several cooperating sessions — cover optimization and certification on 4-core CPU containers, a GPU search fleet on ephemeral cloud notebooks, and delegated agent sessions for audits and sweeps — whose merged artifact history is preserved in the repository. Where a result was produced by one pipeline and re-verified by the other, the text says so, because that independence is load-bearing for trust.

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Chapter 1

Introduction: the least Goldbach summand and the three games

We define the least Goldbach summand $\mathfrak{g}(N)$, survey what is known about it empirically and conjecturally, and set up the three record games — height, threshold, and budget — that organise this thesis. We state the final certified record board, summarise the contributions chapter by chapter, and fix the standard of evidence that every record claim in the thesis must meet.

1.1 Goldbach partitions, and the most lopsided one

Goldbach's conjecture asserts that every even integer $N > 2$ is a sum of two primes; it has been verified for all $N \leq 4 \cdot 10^{18}$ [1]. Heuristically one expects not just one representation but many. The Hardy–Littlewood circle method — which estimates such counts by treating the primes as a random set of the appropriate density and then correcting for divisibility constraints — predicts [3] the following asymptotic. Define

$$r(N) = \#\{p \leq N : p \text{ and } N - p \text{ are prime}\},$$

which counts the two orders separately except on the diagonal. Then

$$r(N) \sim 2\Pi_2 \frac{N}{(\log N)^2} \prod_{\substack{p|N \\ p>2}} \frac{p-1}{p-2}, \quad \Pi_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right). \quad (1.1)$$

Reading this from left to right: $N/(\log N)^2$ is the naive guess, since a number near N is prime with probability about $1/\log N$ (the prime number theorem) and there are N choices for p ; the constant $\Pi_2 \approx 0.6601$ — an infinite product over all odd primes — is the standard correction (the twin-prime constant) that accounts for primes not being truly random modulo small numbers; and the last factor, a product over the odd primes *dividing* N , inflates the count when N is highly composite. The point for us is the heuristic conclusion: $r(N) \rightarrow \infty$, so a typical large even integer has many Goldbach partitions. The interesting question is then not merely whether a representation exists, but how lopsided the most lopsided one must be.

Definition 1.1. For an even integer $N \geq 4$ admitting a Goldbach representation, the *least Goldbach summand* is

$$\mathfrak{g}(N) = \min\{p \text{ prime} : N - p \text{ is prime}\}.$$

This is exactly the quantity $p(n)$ attached to the *minimal Goldbach partition* $N = p + q$ (with p minimal) by Granville, van de Lune and te Riele [4], who conjectured that it never grows faster than a fixed multiple of $(\log N)^2 \log \log N$:

$$\mathfrak{g}(N) = O((\log N)^2 \log \log N). \quad (1.2)$$

The notation $f = O(g)$ means $|f| \leq Cg$ for some constant C and all large N ; (1.2) is a *conjectural* ceiling, not a theorem, and one thing this thesis does is locate it between provable bounds and measured constructive frontiers.

Empirically $\mathfrak{g}$ is tiny and grows very slowly: its record values are catalogued in the OEIS [5] (sequence A020481, with records A025018 and A025019). Exhaustive verification has produced a slowly climbing ladder — among its rungs, $\mathfrak{g} = 1093$ first at $N = 60,119,912$; $\mathfrak{g}(N) \leq 5569$ for all even $N \leq 4 \cdot 10^{14}$ [2]; Herkommer’s tables reach $\mathfrak{g} = 8443$ near $1.2 \cdot 10^{17}$; and the exhaustive-search record is $\mathfrak{g} = 9781$, first attained at $N = 3,325,581,707,333,960,528$, with $\mathfrak{g}(N) \leq 9781$ for *all* even $N \leq 4 \cdot 10^{18}$ [1]. (We reconfirmed the first and last of these deterministically, both being below 2^{64} .) For recent statistical work on the minimal prime in generalised (m_1, m_2) -partitions $n = m_1p + m_2q$, see Juhász–Bartalos [6], observational up to 10^9 . Strikingly, all natural records occur at integers of at most 19 digits. A literature search conducted for this thesis (July 2026) found no published *constructive* records beyond the first-generation constructions of Chapter 2; every large value of $\mathfrak{g}$ reported below was manufactured, and to our knowledge this thesis’s records are the largest and smallest of their kinds that are explicitly reported and independently verifiable anywhere. Our question is:

How large can $\mathfrak{g}(N)$ be forced to be, and at what cost in the size of N ?

1.2 The three record games

There are three natural record questions, and they reward different constructions. The first is the unbounded *height game*: exhibit as large a value of $\mathfrak{g}(N)$ as possible, with no limit on the size of N . An ordinary prime gap is one way to play it, but a strictly wasteful one: a gap must block every integer offset, while a Goldbach desert needs to block only *prime* offsets. Chapter 5 turns that remark into a measured separation.

For comparisons in which size matters, we use two finite games.

Definition 1.2 (Threshold and budget games). For $Q \geq 2$ and $X \geq 4$, define

$$T(Q) = \min\{N : N \text{ even, admitting a Goldbach partition, and } \mathfrak{g}(N) > Q\},$$

$$R(X) = \max\{\mathfrak{g}(N) : N \text{ even, admitting a Goldbach partition, and } N < X\}.$$

The *threshold game* fixes Q and tries to lower the certified upper bound on $T(Q)$; the *budget game* fixes X and tries to raise the certified lower bound on $R(X)$. A companion classical quantity is $S(q)$, the least even N with $\mathfrak{g}(N) = q$ [4]; the threshold function packages many such questions at once, since $T(Q) = \min_{q > Q, q \text{ prime}} S(q)$.

The games can share an incumbent and then separate: a smaller integer just over the threshold improves T but not R ; a higher summand anywhere under the ceiling improves R but not T . Both happened repeatedly in the record history of Chapter 4, which is why the two games are kept distinct throughout.

Throughout the thesis, “digits” means decimal digits, and a *nat* is the natural unit of log-scale accounting: a quantity grows by one nat when it is multiplied by $e \approx 2.718$, so one decimal digit equals $\ln 10 \approx 2.303$ nats. Search costs, size budgets, and probabilities all live on this one scale, which is what makes the economics of Chapters 3–7 possible to state as exchange rates.

1.3 The record board

Table 1.1 states the final certified records of this thesis. Each is an explicit even integer with an exhibited Goldbach partition; the negative side (no smaller prime is a summand) is proved by unconditional witnesses — proper divisors or failed strong probable-prime tests — and the positive side by an elliptic-curve primality certificate for the partition’s large prime, independently re-verified. Chapter 4 describes the constructions and the verification standard; Appendix A maps every row to its machine-checkable artifact.

Table 1.1: The certified record board of this thesis (July 2026), against the first-generation records of the source paper and the natural (exhaustive-verification) frontier.

Game	Incumbent	digits of N	$\mathfrak{g}(N)$
natural (all $N \leq 4 \cdot 10^{18}$)	exhaustive scan record [1]	19	9 781
Height	first generation	237	109 621
Height	this thesis	2 480	1 157 341
Threshold $T(100\,000)$	first generation	199	105 667
Threshold $T(100\,000)$	this thesis	150	104 527
Budget $R(10^{199})$	first generation	199	105 667
Budget $R(10^{199})$	this thesis	199	119 419

Three reading notes on the board. First, the height record $\mathfrak{g} = 1\,157\,341$ exceeds the length of the longest prime gap ever certified with proven endpoints (1 113 106, between primes of 18 662 digits [14]), achieved at 2 480 digits — the construction is not a prime gap in disguise, and Chapter 5 certifies just how far from one it is. Second, the threshold record improved from 199 to 150 digits in six certified rungs across two independent pipelines; the ladder, its cost per rung, and why it stalls near 120–135 digits are the subjects of Chapters 4 and 6. Third, the budget-game value 119 419 is attained by a 199-digit integer and therefore bounds both $R(10^{199})$ and $R(10^{200})$; a separate 200-digit construction with $\mathfrak{g} = 112\,249$ once led the sub- 10^{200} game and is now dominated. No optimality is asserted anywhere: every row is a best-known construction, an upper bound for T and a lower bound for R , and Chapter 6 estimates how far from the truth each one sits (spoiler: very far, provably hard to close).

1.4 Contributions

1. **Foundations and unconditional results** (Chapter 2). Elementary constructions (factorial neighbourhoods with Wilson obstructions; primorial deserts) upgraded, via Dirichlet’s theorem applied twice, to *unconditional* unboundedness of $\mathfrak{g}$ with explicit partitions; the covering-certificate formalism for prime-offset congruence covers, complete and partial; and the transfer of the Ford–Green–Konyagin–Maynard–Tao long-gap covering [11, 12, 13] into the lower bound $\mathfrak{g}(N) \gg \log N \log_2 N \log_4 N / \log_3 N$ along even integers with exhibited partitions.

2. **A validated cost model** (Chapter 3). The expected search cost of a partial-cover construction is e^E with $E = |U| \cdot \text{boost} / \ln N$; the model is calibrated to ± 0.1 nats across five scales and $1.2 \cdot 10^9$ scanned candidates, and it converts record planning from vibes into arithmetic: frontier curves $E(D)$ and $E(Q)$, a digit–compute exchange rate, two binding regimes, and the boost-versus-kills equilibrium that sets cover size.
3. **The certified records** (Chapters 4 and 5). Eleven new certified constructions across three games, including the 150-digit threshold record (the first GPU-scale rung), the 199-digit budget record $\mathbf{g} = 119\,419$, and two height-game constructions ($\mathbf{g} = 1\,157\,341$ at 2 480 digits; $\mathbf{g} = 1\,134\,871$ at 2 692 digits) that beat the certified prime-gap record — the latter with a fully certified local prime gap 126.5 times shorter than its desert. Plus the engineering that made them affordable (measured: sorted testing $1.8\times$, selective testing $3\times$ net, a $430\times$ GPU engine) and a verification standard that survived adversarial audit.
4. **Compression barriers** (Chapter 6). Joint modulus/representative optimization is capped exactly by search entropy (measured to half a digit of its ceiling); set-valued conditioning buys 1–2.8 nats, not orders of magnitude, and the classic cover is the unique freely-enumerable design; oversized covers reduce sub-100-digit records to modular subset-sum with choices at density ≈ 1.026 and to CRT list decoding beyond the Johnson radius — and practical lattice reduction measurably fails to penetrate that radius even on solution-abundant instances. Sub-100 digits is costed at $\sim 9 \cdot 10^5$ GPU-years by the only viable route.
5. **The closure theorem** (Chapter 7). Within a concrete grammar of planted compositeness-certificate mechanisms, congruence covering is optimal: every family is closed by a proof of its own kind (value-set ceilings, cluster bounds, isolation, orbit accounting, a supply-side purchase argument), swept by 13 661 enumerated and 41 adversarially evolved instances with zero passes. The game’s exponent is precisely an integrality gap: the fractional cover is perfect, greedy loses 7.6%, and exact optimization over windows up to half the cover certifies the greedy point locally optimal. Two named conditionals (Hardy–Littlewood independence; no sub-modexp compositeness detection) and the out-of-grammar frontier are stated as open problems.
6. **Methodology** (throughout; collected in Chapter 8). Restart-safe chunked compute with git-committed checkpoints; dual-implementation certification; honest negative results with in-model statistics; and the referee-hardening use of adversarial search — the falsification engine’s first discovery was a $50\times$ accounting error in our own evaluator, and catching it strengthened the theorem it was built to attack.

1.5 Standard of evidence

Every record claim in this thesis separates and discharges three obligations:

- (a) *No small prime is a summand.* For every prime $q < \mathbf{g}(N)$, the complement $N - q$ is proved composite by an unconditional witness: parity, an explicit divisor (from the congruence system or trial division), or a failed strong base-2 probable-prime test — a failed test is a rigorous compositeness proof, since pseudoprimes can only err in the other direction.
- (b) *A Goldbach partition exists.* The partition $N = \mathbf{g} + (N - \mathbf{g})$ is exhibited, with $\mathbf{g}$ proven prime directly (it is always below 2^{64}) and $N - \mathbf{g}$ proven prime by an Atkin–Morain/Goldwasser–Kilian elliptic-curve certificate [17, 18], re-validated by an independent implementation, and

for the CPU-pipeline records additionally by the APR-CL algorithm [19] and a separate full scan.

(c) *The least summand equals the stated value*, which is the conjunction of (a) and (b).

The witnesses are re-derived from the published construction data by standalone verifiers — never trusted from search logs — and the whole package is hash-manifested. Chapter 4 reports the audit history of this standard, including the two occasions a weaker check returned a false PASS and was caught by the remaining legs.

1.6 Notation

$\log_k$ is the k -fold iterated logarithm (so $\log_2 N = \log \log N$, and each extra log grows *extremely* slowly); $\theta(x) = \sum_{p \leq x} \log p$ is Chebyshev’s function; $\pi(x)$ counts primes up to x ; $\gamma \approx 0.5772$ is Euler’s constant; $f \gg g$ means $f \geq cg$ for some constant $c > 0$ and all large arguments (the reverse of $f = O(g)$); “BPSW” denotes the Baillie–PSW probable-prime test [15, 16], “SPRP” a strong probable-prime test to a stated base, and “PRP” a probable prime; “ECPP” is elliptic-curve primality proving. An integer is *y-rough* when it has no prime factor $\leq y$. D denotes a size budget in decimal digits and $B = 10^D$ the corresponding bound; Q a threshold on the least summand; M a Chinese-remainder modulus; U a residual (uncovered) set of prime offsets; and E the search exponent of Chapter 3. Named integers recur across chapters: N_{199} , N_{237} (the first-generation records of Chapter 2), N_{150} (the threshold record), and so on, always with their digit count as subscript.

Chapter 2

Foundations: from factorials to covers, unconditionally

This chapter is the foundations paper, subsuming and updating the July 2026 manuscript Cute Goldbach Gaps. After an elementary factorial warm-up, we prove unconditionally that $\mathfrak{g}$ is unbounded: a primorial obstruction combined with Dirichlet's theorem produces, for every prime P , an even N with an explicit partition and $\mathfrak{g}(N) = \text{nextprime}(P)$. We then formulate the construction as a congruence-cover problem on the prime offsets, in complete and partial forms, prove the realisation theorem that upgrades any cover to an actual Goldbach number, and transfer the Ford–Green–Konyagin–Maynard–Tao long-gap covering into an unconditional lower bound on the maximal order of $\mathfrak{g}$. The chapter closes with the first-generation records this machinery produced — a 1020-digit complete-cover baseline, a 237-digit integer with $\mathfrak{g} = 109\,621$, and a 199-digit integer with $\mathfrak{g} = 105\,667$ — which are the incumbents every later chapter improves.

2.1 Factorial neighbourhoods and Wilson obstructions

Lemma 2.1 (Factorial composites). *For $n \geq 2$ and $2 \leq k \leq n$, the number $n! + k$ is composite. For $n \geq 4$ and $2 \leq k \leq n$, the number $n! - k$ is also composite.*

Proof. In both cases $k \mid n!$, so $k \mid (n! \pm k)$. For the plus side $n! + k > k$, so $n! + k$ is a proper multiple of k . For the minus side, $n! - k = k(n!/k - 1)$ with $n!/k - 1 \geq 2$ exactly when $n! \geq 3k$, which holds for $n \geq 4$; it fails at $n = 3$, $k = 3$, where $3! - 3 = 3$ is prime. $\square$

Lemma 2.2 (Wilson obstructions). *Let $n \geq 2$. (A) If $n + 1$ is prime, then $(n + 1) \mid n! + 1$. (B) If $n + 2$ is prime, then $(n + 2) \mid n! - 1$.*

Recall *Wilson's theorem*: a number $p > 1$ is prime if and only if $(p - 1)! \equiv -1 \pmod{p}$, i.e. the product $1 \cdot 2 \cdots (p - 1)$ is one less than a multiple of p . We use only the forward direction.

Proof. For (A) put $p = n + 1$; Wilson's theorem gives $n! = (p - 1)! \equiv -1 \pmod{p}$. For (B) put $p = n + 2$; then $(p - 1)! = (n + 1)n!$ and $n + 1 \equiv -1 \pmod{p}$, so $-n! \equiv -1$, i.e. $n! \equiv 1 \pmod{p}$. $\square$

Corollary 2.3. *If $n! + 1 > n + 1$ is prime, then $n + 1$ is composite. In particular the factorial primes $n! + 1$ with $n \geq 3$ (for example $n = 3, 11, 27, 37, 41, \dots$) occur only when $n + 1$ is composite.*

2.1.1 The worked example $N = 27! + 28$

Take $N = 27! + 28$, even with 29 digits. Since $27! \equiv 0 \pmod{r}$ for every prime $r \leq 27$,

$$N - q = 27! + (28 - q) \equiv 28 - q \pmod{r} \quad (r \leq 27 \text{ prime}), \quad (2.1)$$

so $N - q$ is killed by a small prime r exactly when $r \mid (28 - q)$: a shifted sieve that eliminates the class $q \equiv 28 \pmod{r}$.

Proposition 2.4. *For $N = 27! + 28$ every prime q with $2 \leq q \leq 67$ has $N - q$ composite, while $N - 71$ is prime; hence $\mathfrak{g}(N) = 71$.*

Proof by explicit witnesses. Table 2.1 gives, for each prime $q \leq 71$, a factor of $N - q$. For $q \leq 23$ the witness divides $28 - q$ via (2.1). Two cases below the winner are notable: $q = 29$ is killed by 29 because $29 = 27 + 2$ is prime, so Lemma 2.2(B) gives $29 \mid 27! - 1 = N - 29$; and $q = 59$ survives the sieve (as $59 - 28 = 31$ is 27-rough) yet $41 \mid 27! - 31 = N - 59$. Finally $71 - 28 = 43$ is 27-rough and $27! - 43$ is prime. $\square$

Table 2.1: Why every small prime fails as a summand of $N = 27! + 28$.

q	$N - q$	factor	q	$N - q$	factor
2	$27! + 26$	2	31	$27! - 3$	3
3	$27! + 25$	5	37	$27! - 9$	3
5	$27! + 23$	23	41	$27! - 13$	13
7	$27! + 21$	3	43	$27! - 15$	3
11	$27! + 17$	17	47	$27! - 19$	19
13	$27! + 15$	3	53	$27! - 25$	5
17	$27! + 11$	11	59	$27! - 31$	41 (generic)
19	$27! + 9$	3	61	$27! - 33$	3
23	$27! + 5$	5	67	$27! - 39$	3
29	$27! - 1$	29 (Wilson)	71	$27! - 43$	prime

By (2.1) a prime q can even be *tested* only if $q - 28$ is 27-rough, so the viable candidates 29, 59, 71, 89, ... are sparse, and 71 is merely the first with a prime complement. Sharper still: $27! + 1$ is prime (Corollary 2.3), the natural bridge $q = 27$, but 27 is composite, so one falls through to 71. For indices that are simultaneously prime and factorial-prime indices ($n = 3, 11, 37, 41$) the bridge works and $\mathfrak{g}(n! + (n + 1)) = n$. Heuristically, the first viable offsets in this factorial family lie on the scale of n , corresponding to roughly $\log N / \log \log N$; the rest of the chapter produces much larger values.

2.2 Primorial deserts, made unconditional

The factorial wastes each small prime on a residue class. Taking N to be a multiple of a primorial wastes nothing.

Lemma 2.5 (Primorial core). *Let P be prime, let $m_P = \prod_{p \leq P} p$ be the primorial — the product of every prime up to P (for instance $m_{11} = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 2310$) — and let N be a multiple of m_P with $N > 2P$. Then $N - q$ is composite for every prime $q \leq P$; i.e. no prime $\leq P$ is a Goldbach summand of N .*

Proof. For a prime $q \leq P$ we have $q \mid m_P \mid N$ (as q is one of the factors of m_P), hence $q \mid N - q$, and $N - q > P \geq q$, so $N - q$ is a proper multiple of q and therefore composite. $\square$

Lemma 2.5 alone gives only claim (a) of the standard of evidence. To reach unboundedness with an *actual* Goldbach number we add Dirichlet’s theorem.

Theorem 2.6 (Unconditional unboundedness). *For every prime P put $s = \text{nextprime}(P)$. There are infinitely many primes $t \equiv -s \pmod{m_P}$, and for any such t the even integer $N = s + t$ satisfies $m_P \mid N$, has the explicit Goldbach partition $N = s + t$, and has*

$$\mathbf{g}(N) = \text{nextprime}(P) > P.$$

Hence $\sup_N \mathbf{g}(N) = \infty$ unconditionally. Moreover the least such t satisfies $\log N \ll P$ by Linnik’s theorem [8], so $\mathbf{g}(N) \gg \log N$ along this family.

Proof. As $\gcd(s, m_P) = 1$ (because $s > P$), Dirichlet [7] gives infinitely many primes $t \equiv -s \pmod{m_P}$. Then $N = s + t \equiv 0 \pmod{m_P}$ and N is even. By Lemma 2.5 no prime $\leq P$ is a summand; the only primes below $s = \text{nextprime}(P)$ are those $\leq P$, so the least summand is at least s . But $N - s = t$ is prime, so s is a summand; hence $\mathbf{g}(N) = s$. Linnik’s theorem bounds the least such prime by $t \ll m_P^L$ for an absolute constant L , and $\log m_P = \theta(P) \sim P$, so $\log N \ll P \sim s = \mathbf{g}(N)$. $\square$

Remark 2.7 (Concrete instances). For $P = 13$ one has $m_{13} = 30030$, $s = 17$, and the least prime $t \equiv -17 \pmod{30030}$ is $t = 30013$, giving $N = 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ with partition $30030 = 17 + 30013$ and $\mathbf{g}(N) = 17$ — all primality deterministic. A larger fully proven example is the primorial $m_{43} = 13082761331670030$ itself: its minimal partition is $m_{43} = 89 + 13082761331669941$ (both proven prime, being $< 2^{64}$), so $\mathbf{g}(m_{43}) = 89$ unconditionally.

Remark 2.8 (The roughness bonus). For the bare choice $N = m_P$, a prime $q > P$ makes $N - q$ coprime to all primes $\leq P$ — the complement is *P-rough* — which improves its chances of being prime, since the small primes that catch most composites are already excluded. By *Mertens’ theorem* (which says $\prod_{p \leq P} (1 - 1/p) \sim e^{-\gamma} / \log P$: the product of $1 - 1/p$ over primes $p \leq P$ shrinks like $1/\log P$), this sieving effect multiplies the heuristic primality probability by $\prod_{p \leq P} (1 - 1/p)^{-1} \sim e^{\gamma} \log P$. The expected least surviving summand then lands near $P(1 + e^{-\gamma}) \approx 1.56 P$. Either way the primorial pays a modulus m_P with $\log m_P = \theta(P) \sim P$ — exponentially large in the target — and the next section removes this waste. The roughness bonus itself, however, is not waste but the engine of everything that follows: it reappears in Chapter 3 as the “boost” factor of the cost model.

2.3 Prime-offset congruence covers

2.3.1 The construction

A primorial spends one prime per summand; a prime-offset congruence cover reuses each prime on a whole residue class. The one structural constraint is parity: a sum of two odd primes is even, so N is kept even and the prime 2 kills only $q = 2$; every *odd* prime below the target must be covered by odd sieving primes.

The strategy itself is classical. Covering systems of congruences were introduced by Erdős precisely to obstruct an additive representation: his construction of an infinite arithmetic progression of odd integers not of the form $2^k + p$ [10] is the direct ancestor of this section (and of the Sierpiński–Riesel tradition that grew from it). What changes here is only the target: the offsets to be blocked are the primes below Q rather than the powers of two.

Proposition 2.9 (Covering certificate). *Let $Q \geq 3$ and let $\{(r, a_r) : r \in \mathcal{R}\}$ be a system of odd primes $\mathcal{R}$ such that every odd prime $q < Q$ has $q \equiv a_r \pmod{r}$ for some $r \in \mathcal{R}$. If N is even, $N \equiv a_r \pmod{r}$ for all $r \in \mathcal{R}$, and*

$$N > Q + \max_{r \in \mathcal{R}} r,$$

then $N - q$ is composite for every prime $q < Q$; i.e. no prime $< Q$ is a Goldbach summand of N .

Proof. For $q = 2$, $N - 2$ is even and > 2 . For an odd prime $q < Q$, pick $r \in \mathcal{R}$ with $q \equiv a_r \equiv N \pmod{r}$; then $r \mid N - q$, and $N - q > (Q + \max_r r) - Q \geq r$, so $r \mid N - q$ properly and $N - q$ is composite. $\square$

Remark 2.10. The size hypothesis only has to make every divisor $r \mid N - q$ proper, for which $N > Q + \max_r r$ suffices; the much stronger $N > Q \prod_r r$ is unnecessary. A common solution N exists modulo $M = 2 \prod_r r$ by the Chinese Remainder Theorem — the classical fact that congruences to pairwise coprime moduli can be satisfied simultaneously, and that the solutions form exactly one residue class modulo the product — and any large enough even representative qualifies.

We build coverings greedily and deterministically.

Algorithm (greedy covering of the odd primes below Q).

1. Let $T = \{\text{odd primes } < Q\}$ and require $N \equiv 0 \pmod{2}$.
2. For $r = 3, 5, 7, 11, \dots$ in increasing order, while $T \neq \emptyset$: let a_r be a residue mod r containing the most elements of T (smallest such residue on ties), require $N \equiv a_r \pmod{r}$, and delete from T every $q \equiv a_r \pmod{r}$.
3. Recover N from the congruences by the Chinese Remainder Theorem.

The resulting modulus is $M = 2 \prod_{r \in \mathcal{R}} r$ — twice the product of the sieving primes actually used. For the computed targets it is far smaller than the primorial $\prod_{p < Q} p$, because each sieving prime is reused across an entire residue class rather than assigned to one offset. (Chapter 4 replaces this naive greedy pass with weighted set cover plus annealing; Chapter 7 asks, and largely answers, how far from optimal such covers are.)

2.3.2 Partial covers and progression search

A complete cover is convenient to verify, but it is not the cheapest way to make N small. Near the end of the greedy algorithm, a new sieving prime may remove only one or two remaining offsets while still multiplying the modulus by r . That is an expensive way to buy a single compositeness proof.

The alternative is to stop early. Let U be the odd prime offsets below Q not covered by the chosen congruences — the *residual set*. CRT gives one residue N_0 modulo

$$M = 2 \prod_{r \in \mathcal{R}} r,$$

so every candidate has the form

$$N = N_0 + kM, \quad k = 0, 1, 2, \dots \tag{2.2}$$

The fixed congruences dispose of most offsets for every k . For each remaining $q \in U$, we test $N - q$ directly. A failed probable-prime test is a rigorous compositeness result: pseudoprimes can create false positives, but they cannot turn a prime into a reported composite. Thus a fast one-sided test is safe as a search filter, while the final surviving complement still needs a primality certificate.

Proposition 2.11 (Partial-cover certificate). *Let U be the odd primes $q < Q$ not covered by the congruences of Proposition 2.9. Let N be a sufficiently large even solution of those congruences. If $N - q$ is composite for every $q \in U$, then no prime $q < Q$ is a Goldbach summand of N .*

Proof. The congruence witnesses prove compositeness for every covered odd prime; parity handles $q = 2$; and the hypothesis handles the residual set U . $\square$

Most values of k fail the hypothesis — some residual complement turns out to be prime, which is not a disaster but a small Goldbach partition, i.e. a losing lottery ticket — and the construction becomes a *search* along the progression (2.2). What that search costs, and how predictably, is the subject of Chapter 3; it is the single question the rest of the thesis orbits.

2.3.3 Realising a covering unconditionally

Proposition 2.9 still only gives claim (a). The same Dirichlet idea as in §2.2, applied twice, upgrades any covering to an actual Goldbach number.

Theorem 2.12 (Unconditional realisation). *Let $Q \geq 3$ and let $\{(r, a_r) : r \in \mathcal{R}\}$ cover the odd primes below Q , where every $r \in \mathcal{R}$ is an odd prime. Then there are infinitely many even integers $N > Q + \max_{r \in \mathcal{R}} r$ admitting an explicit Goldbach partition $N = s + t$ with $N \equiv a_r \pmod{r}$ for all $r \in \mathcal{R}$. Every such N has $\mathbf{g}(N) \geq Q$.*

Proof. For each $r \in \mathcal{R}$ choose $b_r \in \{1, \dots, r-1\} \setminus \{a_r\}$, and also require $s \equiv 1 \pmod{2}$. The CRT gives a reduced residue class b modulo $2 \prod_r r$. Dirichlet gives infinitely many primes in that class; call one of them s . If $s < Q$, then s is an odd covered prime, so $s \equiv a_r \pmod{r}$ for some r , contradicting $s \equiv b_r \not\equiv a_r \pmod{r}$. Hence $s \geq Q$.

Next require

$$t \equiv a_r - s \pmod{r} \quad (r \in \mathcal{R}), \quad t \equiv 1 \pmod{2}.$$

Every residue $a_r - s$ is nonzero modulo r , so this is a reduced residue class modulo $2 \prod_r r$. Dirichlet gives infinitely many prime values of t ; keep those large enough that $s + t > Q + \max_r r$, and put $N = s + t$. Then N is even, $N \equiv a_r \pmod{r}$, and $N = s + t$ is an explicit Goldbach partition. Proposition 2.9 excludes every prime below Q , so $\mathbf{g}(N) \geq Q$. $\square$

Remark 2.13 (Quantitative form for initial-segment covers). The covering used in Theorem 2.15 below has one modulus for every prime $p \leq x$. In that setting the two residue classes above are reduced modulo $M = 2 \prod_{p \leq x} p$. Linnik's theorem [8] supplies primes $s, t \ll M^L$ for an absolute constant L . Their residues are nonzero modulo every prime at most x , and the covering property forces the odd prime s to satisfy $s \geq Q$. Consequently the resulting integer satisfies

$$\log N \ll \log M = \theta(x) \ll x.$$

This quantitative statement is used only for those initial-segment covers; no such bound is asserted for an arbitrary sparse finite cover.

2.4 Transfer from large prime gaps

Proposition 2.14. *If $[N - Q, N - 2]$ contains no prime, then no prime $q \leq Q$ is a Goldbach summand of N . Consequently, if N admits a Goldbach representation, then $\mathbf{g}(N) > Q$.*

Proof. Every prime $q \in [2, Q]$ has $N - q \in [N - Q, N - 2]$, hence $N - q$ is composite. $\square$

So a prime gap of length $\geq Q$ just below N forces $\mathfrak{g}(N) > Q$: *prime gaps transfer to least Goldbach summands*. The transfer is, however, strictly one-directional. Forcing $\mathfrak{g}(N) > Q$ requires $N - q$ composite only for *prime* offsets $q < Q$, not for every integer in an interval; the covering of Proposition 2.9 covers just the primes below Q , a far sparser target. Thus “large least Goldbach summand” is a relaxation of “large prime gap,” and one should not expect the two problems to coincide. Chapter 5 measures the separation on certified examples; here we exploit the easy direction.

Writing $G(X)$ for the largest gap between consecutive primes below X , the work of Ford–Green–Konyagin–Tao and of Maynard [11, 12], sharpened in [13], gives

$$G(X) \gg \frac{\log X \log_2 X \log_4 X}{\log_3 X},$$

improving Rankin [9]. In words: there are stretches of consecutive composite integers below X of length at least a constant times $\log X \cdot \log_2 X \cdot \log_4 X / \log_3 X$. The dominant factor is $\log X$ (the average prime gap near X); the three iterated-log factors are a slowly growing extra boost, and since each $\log_k$ grows glacially this is only a little more than $\log X$. Feeding the covering that underlies these bounds into Theorem 2.12 (which costs only a constant factor in $\log N$, by Linnik) makes the transfer unconditional:

Theorem 2.15. *There are infinitely many even integers N , each with an explicit Goldbach representation, such that*

$$\mathfrak{g}(N) \gg \frac{\log N \log_2 N \log_4 N}{\log_3 N}.$$

Proof. The Erdős–Rankin method, in the sharpened form of Ford–Green–Konyagin–Maynard–Tao, supplies the covering. Precisely, [13] shows (in the standard covering formulation of the method, the quantity $Y(x)$ of [11, 13]) that there is an absolute constant $c > 0$ such that for all large x one may choose residue classes $a_p \pmod{p}$, one for each prime $p \leq x$, whose union contains every integer in $[1, y]$, where

$$y = \left\lfloor c \frac{x \log x \log_3 x}{\log_2 x} \right\rfloor;$$

the prime-gap bound is deduced from exactly this covering by the Chinese Remainder Theorem.

Normalisation. If a_2 is odd, replace every class a_p by $a_p - 1$; the shifted system covers $[0, y - 1] \supseteq [1, y - 1]$. So, at the cost of replacing y by $y - 1$, we may assume $2 \mid a_2$. Then no odd integer in $[1, y - 1]$ is covered by the class modulo 2, so every odd integer $m \leq y - 1$ — in particular every odd prime $q < Q := y - 1$ — satisfies $q \equiv a_r \pmod{r}$ for some *odd* prime $r \leq x$. The pairs (r, a_r) over the odd primes $r \leq x$ therefore form a covering of the odd primes below Q in the sense of Proposition 2.9: it covers far more (all odd integers below Q), but only the primes are needed, and self-covering ($r = q, a_r = 0$) is harmless, exactly as in the computed systems of Chapter 4.

Realisation. Theorem 2.12 applied to this covering yields infinitely many even N with an explicit partition and $\mathfrak{g}(N) \geq Q$; by Remark 2.13 we may take

$$\log N \ll \log 2 + \theta(x) \ll x.$$

Conclusion. Let C be the absolute constant with $\log N \leq Cx$. For large u the function $u \mapsto u \log u \log_3 u / \log_2 u$ is increasing, and each iterated logarithm of $\log N/C$ is $(1 + o(1))$ times

the corresponding iterated logarithm of $\log N$; hence

$$\begin{aligned} \mathfrak{g}(N) \geq Q &\gg \frac{x \log x \log_3 x}{\log_2 x} \geq \frac{\frac{\log N}{C} \log\left(\frac{\log N}{C}\right) \log_3\left(\frac{\log N}{C}\right)}{\log_2\left(\frac{\log N}{C}\right)} \\ &\gg \frac{\log N \log_2 N \log_4 N}{\log_3 N}. \end{aligned}$$

For each sufficiently large x , select one realisation N_x satisfying this Linnik bound. Since $\mathfrak{g}(N_x) \geq Q(x) \rightarrow \infty$, the selected integers are unbounded and therefore contain infinitely many distinct values. This proves the claim. $\square$

Together with the Granville–van de Lune–te Riele conjecture (1.2), this brackets the maximal order of $\mathfrak{g}$ between a logarithmic-type lower bound and a $(\log N)^2$ -type conjectural ceiling:

$$\underbrace{\frac{\log N \log_2 N \log_4 N}{\log_3 N}}_{\text{rigorous lower-bound scale}} \ll \max_{N' \leq N} \mathfrak{g}(N') \lesssim \underbrace{(\log N)^2 \log \log N}_{\text{conjectural ceiling}}.$$

The explicit constructions of this thesis do not settle that gap; their value is that they turn the problem into auditable objects — a congruence list, a CRT progression, a finite residual set, and a primality certificate — and, in Chapters 6 and 7, into measured statements about which parts of the gap are closable at all. We note one reappearance of this asymptotic machinery: the FGKMT covering enters again in Chapter 7 not as a lower bound but as the one known *rounding technology* that might beat greedy covers, which is this thesis’s central open problem.

2.5 The first-generation records

The machinery above produced the first constructive records in this problem family, which the rest of the thesis treats as its baseline. All three claims (a)/(b)/(c) were discharged to the standard of Chapter 1, and all were re-verified inside the thesis pipeline before any successor record was accepted.

The complete-cover baseline. A complete greedy cover of the odd primes below $Q = 10^5$ uses 356 congruences and forces a modulus of 1020 digits; its least even representative has $\mathfrak{g}(N) = 100\,747$. Every excluded offset has a small factor visible directly from the congruence system, so verification is little more than modular arithmetic — a *proof-oriented build* that overpays roughly sevenfold in digits for certainty it does not need. (A complete cover at $Q = 10^8$ was also computed as a scaling probe: 327 455 digits, with no partition exhibited.)

The hybrid ablation. Stopping the cover early and searching the progression (2.2) shrinks the integer dramatically. The engineering path, at fixed target $Q = 10^5$:

The 237-digit height record. The final row is the first-generation height incumbent: a 237-digit even integer $N_{237} = N_0 + 299\,948\,M$ over a partial cover of 102 odd congruences plus parity (M of 232 digits, largest modulus 1217), with

$$\mathfrak{g}(N_{237}) = 109\,621.$$

Every prime offset below 109 621 is excluded — 9 691 by explicit congruence divisors and 731 by strong base-2 compositeness witnesses — and the complement $N_{237} - 109\,621$ is ECPP-certified prime.

Table 2.2: First-generation ablation at target $Q = 10^5$. “Congr.” includes parity; “residual” counts prime offsets below 10^5 left for direct testing; the last column is the first prime offset whose complement is (certified) prime.

Method	Congr.	Digits M	Residual	k	Digits N	$\mathbf{g}(N)$
Complete greedy cover	356	1020	0	0	1020	100 747
Prefix partial cover	168	416	385	74	418	101 599
Coordinate descent	110	248	580	1 751	251	100 981
Sparse weighted cover	103	232	610	299 948	237	109 621

The 199-digit dual incumbent. Re-optimising the cover for the strict budget $N < 10^{199}$ gives a 199-digit integer $N_{199} = N_0 + 876\,554\,M$ (89 odd congruences plus parity, M of 193 digits) with

$$\mathbf{g}(N_{199}) = 105\,667,$$

proved by 9 335 parity-or-congruence witnesses, 744 strong base-2 witnesses, and a 34-step elliptic-curve certificate for the complement. It entered the thesis as the incumbent for *both* bounded games: $T(100\,000) \leq N_{199}$ and $R(10^{199}) \geq 105\,667$.

These records made the covering paradigm concrete, and they framed the questions the thesis then had to answer. Why 237 digits and not 150? Is the greedy cover any good? What does a progression search actually cost, and can that cost be predicted rather than experienced? The next chapter builds the measuring instruments; Chapter 4 spends them.

Chapter 3

The cost model: what a desert costs, and how predictably

A partial-cover construction is a lottery: each progression element survives only if every residual complement happens to be composite. This chapter turns that lottery into a measured science. The success density is e^{-E} with $E = |U| \cdot \text{boost} / \ln N$; we derive the model, refine it with the exact survivor-count distribution, and validate it to ± 0.1 nats across five scales and $1.2 \cdot 10^9$ scanned candidates — the dice are exactly as loaded as predicted, every time. On top of the model we build the economics: the digit-budget identity that splits $\ln N$ into modulus nats and search nats, the digit-compute exchange rate, the two binding regimes of the bounded games, the boost-versus-kills equilibrium that sets cover size, and the frontier curves $E(D)$ and $E(Q)$ that price every record before it is attempted. Two consequences frame the rest of the thesis: a million-desert below 200 digits exists in abundance heuristically yet costs 10^{85} primality tests to exhibit by covers — a construction-versus-existence gap of fifty-nine orders of magnitude beyond cosmological — and the calibrated truth for $T(100\,000)$ sits near 10^{53} , a floor our 150-digit record approaches only in logarithm.

3.1 The lottery

Fix a partial cover: odd prime moduli $\mathcal{R}$ with residues a_r , parity, the modulus $M = 2 \prod_{r \in \mathcal{R}} r$, and the residual set U of uncovered odd primes below the target Q . Every candidate is $N = N_0 + kM$, and k wins exactly when all $|U|$ residual complements $N - q$ ($q \in U$) are composite (Proposition 2.11). Losing tickets are cheap to recognise: scan the residual complements and stop at the first probable prime. Winning tickets are certified as in Chapter 1.

Two structural facts make the lottery analysable. First, for residual q the complement $N - q$ is automatically coprime to 2 and to every cover modulus: $q \not\equiv a_r = N \pmod r$ means $r \nmid N - q$. The cover therefore pre-sieves its own residuals — the roughness bonus of Remark 2.8 in its useful form. Second, the residual complements at a given k are distinct integers of size $\approx N$ whose only shared structure is exactly that coprimality (a fact made precise, and load-bearing, in Chapter 7).

3.2 The density model

A random integer near N is prime with probability about $1/\ln N$. Conditioning on coprimality to a set S of primes divides the candidate space by $\prod_{p \in S} (1 - 1/p)$, so the conditional primality

probability for one residual complement is

$$p_1 = \frac{\text{boost}}{\ln N}, \quad \text{boost} = 2 \prod_{r \in \mathcal{R}} \frac{r}{r-1},$$

the factor 2 from oddness. When the cover contains every prime up to $r_{\max}$, Mertens’ theorem gives $\text{boost} \approx 2e^\gamma \ln r_{\max}$; real covers skip some primes and the product is computed exactly. Treating the $|U|$ residual primality events as independent (the whole content of that assumption is examined in Chapter 7; empirically it holds to ≤ 0.1 nats), the probability that a candidate wins is

$$\prod_{q \in U} (1 - p_1) \approx e^{-|U|p_1} = e^{-E}, \quad \boxed{E = \frac{|U| \cdot \text{boost}}{\ln N}}$$

and the expected number of progression elements to the first hit is e^E . The exponent E is the single number that decided every campaign in this thesis: it is the negative log of the per-candidate success density, measured in nats.

The exact refinement. In practice the search engine sieves each block of k by all primes up to a depth B_{sieve} (beyond the cover moduli), so at each k only $a_k \leq |U|$ complements remain *alive*, and a surviving complement is prime with the deeper conditional probability $\hat{p}$. The exact per- k density is $(1 - \hat{p})^{a_k}$, and the honest exponent is

$$E_{\text{exact}} = -\ln \mathbb{E}_k[(1 - \hat{p})^{a_k}],$$

the expectation over the sieve’s survivor-count distribution (measured mean ≈ 316 , standard deviation ≈ 14 for a typical $|U| = 750$ cover at 199 digits, with $\hat{p} \approx 0.059$). Because the average of an exponential exceeds the exponential of the average, the fluctuation in a_k is worth a small bonus, $e^{\hat{p}^2 \sigma_a^2 / 2} \approx 0.2\text{--}0.3$ nats; the naive formula undercounts by exactly that much, and the measured densities below confirm the exact form. The same survivor counts drive the search-order optimizations of Chapter 4.

3.3 Validation: five scales, $1.2 \cdot 10^9$ candidates

The model was validated before it was trusted, at every scale it was later used, by comparing the predicted exponent against $\hat{E} = -\ln(\text{measured survival rate})$ on fully logged scans.

Table 3.1: Density-model validation. “Predicted” is E (exact form where computed); “measured” is $\hat{E}$ from the logged scan. The two toy deserts are small constructions built to be exhaustively scannable.

construction	digits of N	predicted E	measured $\hat{E}$
toy desert	26	12.53	12.51
toy desert	62	13.52	13.57
dual-game record cover	197	18.20	18.2
threshold record cover	195	17.51	17.4
GPU threshold cover	150	24.91	(one hit at $\lambda = 0.63$)
mega-desert cover	2692	6.60	6.56

Across three fully instrumented campaign covers the running discrepancy $|\hat{E} - E|$ stayed below 0.1 nats over $1.2 \cdot 10^9$ scanned progression elements, with per-slice survivor rates within $\pm 2\%$ of

prediction across more than 500 logged slices. The 150-digit row deserves its asterisk explained: that campaign scanned its entire k -round $[0, 7.5 \cdot 10^{10}]$ end to end ($\sim 2.1 \cdot 10^{12}$ Fermat tests) and collected exactly one hit against a Poisson mean of 0.63 — an unremarkable draw, and the model stayed unfalsified at the largest scale ever pushed. This validation is what licenses every forecast in the thesis: when we say a rung costs $e^{27.3}$ candidates, that number has survived contact with 10^9 -scale reality repeatedly.

We flag the honest converse: the validation also *bounds* every dream of beating the lottery. Any exploitable correlation between residual complements, any hidden compositeness bias in constructed N , would show up as $\hat{E} < E$; over $1.2 \cdot 10^9$ candidates nothing beyond 0.1 nats ever did. Chapter 7 explains why not.

3.4 The digit-budget identity and the exchange rate

Take logarithms of $N \approx kM$:

$$\underbrace{\ln N}_{\text{record size (nats)}} = \underbrace{\ln M}_{\text{modulus nats}} + \underbrace{\ln k}_{\text{search nats}} .$$

Modulus nats are spent at design time on congruences (each modulus r costs $\ln r$ and kills a residue class of offsets); search nats are spent at run time scanning the progression ($\ln k^* \approx E$ at the expected hit). The two currencies buy the same digits, and the whole craft of the bounded games is arbitraging them:

- **Search nats cost only compute.** A 4-core CPU pipeline demonstrated $\ln k \approx 20$; the GPU fleet of Chapter 4 afforded $\ln k \approx 25$. Throughput converts to record digits at the model’s slope — which is why the single largest threshold-game jump in this thesis ($179 \rightarrow 150$ digits) was purely an engineering event: the CUDA engine’s $430\times$ throughput is $\ln 430 \approx 6.1$ nats, precisely the exponent gap ($E = 24.9$ vs ≈ 19) between the two rungs.
- **The exchange rate is the frontier slope.** Near 200 digits, $dE/dD \approx -0.10$ to -0.19 per digit (so $10\times$ compute buys 12–23 digits there); by 150 digits the slope is ≈ -0.32 , and near 100 digits ≈ -0.5 ($10\times$ buys only ~ 5). The certified ladder measured it directly: each ~ 7 -digit rung of the $199 \rightarrow 179$ descent cost $\approx 2.5\times$ the previous search.
- **Cover quality is the third currency.** At fixed $\ln M$, better residue choices shrink $|U|$; each 1% of $|U|$ is worth $e^{0.01 \cdot E}$ of search. Chapter 7 shows the greedy covers are locally perfect and globally within an integrality gap worth at most a few nats — the least solved, most valuable margin in the game.

3.5 Two binding regimes, and the equilibrium that sizes covers

Compute-bound versus range-bound. With no size cap, the only constraint is affordable search: pick the cover minimising $\ln M + E$ and scan. Under a cap $N < B$ (the bounded games), the progression only holds $k_{\max} = B/M$ candidates, and the binding constraint can flip from compute to *range*. In the range-bound regime the optimum counterintuitively *shrinks* the modulus below its unconstrained value: each nat removed from $\ln M$ buys a full nat of k -range while costing only ≈ 0.2 nats of exponent (the frontier slope in disguise). This was derived, then confirmed twice in campaign post-mortems: raw GPU throughput pointed at a 200-digit budget target without re-optimising M failed to beat a tuned CPU cover, and the tuned cover’s own success confirmed the shrunken-modulus optimum.

The boost-versus-kills equilibrium. What stops a cover from growing forever? Adding a modulus r with its best residue class kills some number of residuals (kills $\approx |U|/(r-1)$ on random structure, less at the picked-over endgame), lowering E by kills $\cdot$ boost/ $\ln N$; but it also multiplies the boost by $r/(r-1)$, *raising* every surviving residual’s primality odds and hence E by $|U| \cdot$ boost/ $((r-1)\ln N)$. The two cancel at

$$\text{kills} \approx \frac{|U|}{r-1},$$

and past that point new moduli are net-zero or worse. Measured at $B = 10^{200}$, $Q = 122\,000$: adding $r = 461$ to the standing 88-modulus cover kills 3.5 residuals (-0.084 nats) and boosts the survivors ($+0.082$ nats) — net $+0.00$. Two consequences. First, at large budgets the cover is *not* budget-limited ($\ln M$ used 437 of ~ 458 available nats in that instance); it sits at the equilibrium, which is why leftover digit budget buys nothing and why the frontier is flat against re-derivation — independent greedy runs at the same Q land within 10^{-3} nats of each other. Second, the binding constraint crosses over: at $Q = 122\,000$ under 199 digits the *budget* binds, while at $Q = 200\,000$ the *equilibrium* binds. Both regimes were mapped before the campaigns that used them.

3.6 The frontier curves

Everything above compresses into two curves, computed by building actual covers (lazy-greedy, digit-capped, annealed) and evaluating E exactly.

Table 3.2: The digit frontier $E(D)$ at fixed threshold $Q = 100\,003$ (greedy digit-capped covers). “Candidates” is the expected search e^E .

D (digits)	moduli	$ U $	E	candidates
100	52	973	41.3	$10^{17.9}$
120	60	899	32.8	$10^{14.2}$
130	64	862	29.5	$10^{12.8}$
140	68	830	26.8	$10^{11.6}$
150	72	802	24.4	$10^{10.6}$
160	76	773	22.3	$10^{9.7}$
170	79	754	20.6	$10^{8.9}$
195	89	692	16.8	$10^{7.3}$

Table 3.3: The threshold frontier $E(Q)$ at fixed budget 199 digits (same 88-modulus digit budget, greedy + coordinate ascent + kicks; boost 11.01). The last column prices each rung against its predecessor.

Q	$ U $	E	candidates	$\times$ previous
122 000	867	20.83	$1.1 \cdot 10^9$	—
135 000	1 004	24.12	$3.0 \cdot 10^{10}$	$\times 27$
150 000	1 138	27.34	$7.5 \cdot 10^{11}$	$\times 25$
175 000	1 335	32.07	$8.5 \cdot 10^{13}$	$\times 113$
200 000	1 569	37.69	$2.4 \cdot 10^{16}$	$\times 280$

Two planning corollaries. Because rung spacing is geometric, climbing *every* rung of a ladder costs only $\approx 4\%$ more than jumping straight to the top — and banks a certified record, plus a fresh

calibration of $\hat{E}$, at each step. And each curve exposes its regime: Table 3.2 is the compute-bound threshold game (digits fall as affordable E rises), Table 3.3 the range-bound budget game (the target rises until the equilibrium chokes it).

3.7 Construction versus existence: the million-desert study

How far can the paradigm be pushed? As a stress test we priced the *million desert*: an even $N < 10^{200}$ with $\mathfrak{g}(N) > 10^6$, i.e. all 78 498 primes $q \leq 10^6$ blocked at once. Note the conjectural ceiling (1.2) evaluates to $(\ln N)^2 \ln \ln N \approx 1.3 \cdot 10^6$ exactly at 200 digits: this target sits *at* the extremal constant of the Granville–van de Lune–te Riele law, not beyond it. And existence is not the issue: for unstructured even N near 10^{200} the all-composite exponent is $E_{\text{random}} \approx 445$, so the expected count of qualifying integers below 10^{200} is $10^{200} e^{-445} \approx 10^{6.5}$ — they exist in abundance, heuristically.

Exhibiting one is another matter. The frontier $E(D)$ at $Q = 10^6$:

Table 3.4: The million-desert frontier (lazy-greedy digit-capped covers, $Q = 10^6$).

D (digits)	moduli	$ U $	E	search cost e^E
199	91	8 157	196.7	$10^{85.4}$
300	127	7 311	124.3	$10^{54.0}$
400	161	6 728	89.3	$10^{38.8}$
600	226	5 865	54.9	$10^{23.8}$
800	288	5 241	38.2	$10^{16.6}$
1000	347	4 741	28.4	$10^{12.3}$
1300	435	4 100	19.6	$10^{8.5}$
1600	519	3 588	14.2	$10^{6.2}$
2400	736	2 492	6.9	$10^{3.0}$

The 2400-digit row reproduces the regime of the certified mega-desert of Chapter 5 (E 6.9 vs its measured 6.56) — the ladder is consistent end to end. Reading off feasibility: a record-scale search ($1.4 \cdot 10^8$ candidates) needs $\sim 1\,470$ digits; a heavy single box (10^9) $\sim 1\,250$; a distributed fleet ($7 \cdot 10^{10}$) $\sim 1\,120$; and a cosmological budget (10^{12} tests per second for the age of the universe, $E \leq 68$) still needs ~ 520 digits. A verified 1 250-digit blueprint (420 moduli, $|U| = 4\,205$, $E = 20.74$, about 235 core-days) is committed in the artifact repository as the realistic version of this target.

At 199 digits the wall is structural. The best 199-digit system found — 91 moduli, residue-annealed — leaves $|U| = 8\,146$ and $E = 196.4$; annealing recovers only $\sim 0.3\%$ of E . Structural in the sieve sense, not the counting sense: the 91 best classes hold 148 290 slots for 78 497 targets, a $1.89\times$ supply, so a union bound alone forbids nothing — but prime residue classes overlap like independent sieves, coverage stalls near 90%, and 60 000 anneal moves reclaim 11 residuals out of 8 157. Joint optimization of representative and coverage (Chapter 6) moves E from 196 to ≈ 194 . So: about $10^{6.5}$ million-deserts exist below 10^{200} , and the cheapest known path to *point at one* costs 10^{85} primality tests — a construction-versus-existence gap of fifty-nine orders of magnitude beyond a cosmological compute budget. That gap, in miniature, is also the story of the sub-100-digit threshold game (Chapter 6); the million-desert study is the thesis’s cleanest exhibit that *knowing something exists and being able to exhibit it are different sciences*.

3.8 Poisson discipline: negative results without excuses

The lottery is memoryless: conditioned on no hit so far, the expected remaining wait is unchanged. That single fact, taken seriously, is most of campaign discipline.

- **Spread.** Over the $n = 7$ fully-logged searches that ended in hits, first hits landed at $0.25\times$ – $3.0\times$ the expected e^E (median $0.5\times$) on a consistent consumed-candidates basis. The exponential clock is merciless about variance: plan budgets at $3 \times E$, in writing, before starting.
- **Misses are data.** Campaign 1 of the PhD track retired hitless at $3.18 \cdot 10^8$ elements — survival probability 0.30 under the calibrated model, an unlucky but ordinary draw; per-slice $\hat{p}$ and survivor counts matched prediction throughout, ruling out implementation error. Campaign 2 stopped at $5.6 \cdot 10^8$ elements ($P \approx 0.22$). The model was never adjusted after the fact to absorb a miss; it never needed to be.
- **The tail teaches.** The first-generation-beating 199-digit dual record (Chapter 4) hit at 99.9% of its committed range, $3.0\times$ the selection-adjusted expectation — the outlier that taught the $3\times$ rule.
- **Ratchet, don’t grand-slam.** Under fixed compute, a ladder of certified rungs dominates a single deep campaign: geometric rung spacing makes the total cost $\approx 1.04\times$ the top rung (§3.6), each rung banks a record and a calibration, and optimal stopping becomes arithmetic. The thesis learned this the expensive way — a breadth-first 400-variant campaign capped its best possible outcome and exhausted hitless while a rival single-progression depth-first design (unbounded runway, bad luck costs digits rather than termination) banked the record. Both post-mortems are in Chapter 4.

3.9 Calibrating the truth: how far are the records from $T(Q)$ itself?

The exhaustive data fix the constant in the conjectural law (1.2): $\mathfrak{g} \leq 9\,781$ for all even $N \leq 4 \cdot 10^{18}$ gives $C \approx 1.4$ in $\mathfrak{g}_{\max}(X) \approx C \ln^2 X \ln \ln X$. Inverting at $Q = 10^5$ puts the *true* $T(100\,000)$ near 10^{53} — plus-or-minus a few in the exponent from the unknown constant and residue-luck tails — i.e. a 53-digit integer, unreachable by any imaginable exhaustive verification (it would take $\sim 10^{52}$ scans) and, Chapter 6 will argue, essentially unreachable by construction too. Our 150-digit record is thus about $2.9\times$ the truth *in logarithm*. For the budget game the truth is closer: $R(10^{200}) \approx C \cdot (460.5)^2 \ln(460.5) \approx 1.8$ million, against a certified 119 419 — and there the gap is compute-and-covers, not existence.

For threshold games the model collapses to a useful pair of envelope identities (cover-quality factor $\varphi \approx 0.60$ for current optimizers; $\ln \mathcal{C}$ the affordable search exponent):

$$\begin{aligned} \text{digits}(N) &\gtrsim \frac{0.57 \varphi \pi(Q)}{\ln \mathcal{C}} && \text{(compute-bound wall),} \\ \text{digits}(N) &\gtrsim \frac{\sqrt{c \pi(Q)}}{2.3}, \quad c \in [0.8, 1.3] && \text{(absolute floor),} \end{aligned}$$

the floor being the regime where the “construction” has degenerated into nature’s own ensemble scan — which is why the floor (38–49 digits at $Q = 10^5$) and the truth (~ 53 digits) nearly coincide. Everything between the frontier tables and that floor is the subject of Chapters 6 and 7: how much is buyable with compute, how much with better covers, and how much is provably out of reach.

Chapter 4

Engineering the records: two pipelines, one ladder

This is the records paper. Two independent pipelines — a 4-core CPU stack for cover optimization, search, and certification, and a GPU fleet on ephemeral cloud notebooks for raw scanning — drove the bounded-game records from the first-generation 199 digits down to 150 for the threshold game $T(100\,000)$, and up to $\mathfrak{g} = 119\,419$ under 10^{199} for the budget game. We describe the cover optimizer, the search engine and its three measured optimizations (survivor-ordered testing, $1.8\times$; selective testing, $3\times$ net; sieve-depth tuning), the checkpoint discipline that survived roughly ten container losses without losing a byte of coverage, and the five-leg certification standard that survived an adversarial audit — including the two occasions a weaker check returned a false PASS and was caught. Every certified construction is tabulated with its cover parameters and search cost; the negative campaigns are reported with the same care, because their statistics are the model validation of Chapter 3 in action.

4.1 Two pipelines

The records were produced by two deliberately independent stacks, developed in parallel sessions against the same artifact repository.

- **The CPU pipeline** runs on 4-core Linux containers: weighted-set-cover construction with GRASP and simulated annealing, CRT progression search with a numpy block presieve and gmpy2 strong-probable-prime tests (about $6 \cdot 10^3$ candidates per second end to end), and the full certification stack — PARI/GP ECPP [22, 18], an independent pure-Python certificate checker, APR-CL [19] as a second algorithm, and a separate full-scan crosscheck.
- **The GPU pipeline** is a CUDA engine on rotating cloud notebook sessions: a bit-matrix sieve and Montgomery-CIOS base-2 Fermat waves, validated bit-for-bit against the Python implementation, at 2.5 million tests per second on a T4 and 8.5 million on an A100 — about $430\times$ the 4-core pipeline. A fleet driver shards the k -range across sessions and checkpoints progress in git.

The division of labour follows the currencies of §3.4: covers and certificates are latency-tolerant and optimization-heavy (CPU); the scan is embarrassingly parallel modular exponentiation (GPU). A shared JSON spec schema (cover pairs, N_0 , M , k -ranges) is the handoff interface, and every hit found by either pipeline is certified by both verification stacks before it is called a record.

4.2 Building covers

The greedy pass of Chapter 2 is the seed, upgraded three ways.

Weighted set cover. Choosing modulus r costs $\ln r$ nats of modulus budget; a candidate pair (r, a_r) is scored by newly covered offsets per nat. Lazy-greedy selection by this ratio, rather than raw coverage, is what turned the 1020-digit complete cover into the 232-digit partial cover of the first generation.

Local refinement. A coordinate-descent pass revisits each chosen modulus and re-picks its residue against the current uncovered set, to a fixpoint; GRASP-style randomized ruin-and-recreate with simulated-annealing acceptance explores further. The refined objective is $\ln M + E$ — modulus nats plus search nats, the two currencies on one scale. Two robust findings: annealing from multiple seeds reliably reproduces the greedy-plus-descent cover (zero improvement, a plateau that Chapter 7 explains and largely certifies), and the frontier is flat — independent re-derivations at the same Q and budget match E to three decimals.

Variant catalogs. A single progression under a size cap holds only $k_{\max} = B/M$ candidates. Residue-swap *variants* — same moduli, alternative residues of near-equal coverage — give independent progressions at $\approx$ zero design cost. The plateau implies many exactly-equal assignments exist: measured on the $Q = 100\,003$ threshold cover, 26 zero-penalty single swaps, 325 pairs, and roughly 2 600 triples, a catalog of $\sim 3\,000$ progressions at exactly the base $|U|$. Catalogs are what made the range-bound campaigns possible at all.

An adversarial aside. An external evolutionary-search service (AlphaEvolve) was pointed at the cover-step optimizer twice before this thesis’s own closure work adopted it properly (Chapter 7). The first run’s fitness was unconstrained — $\text{digits}(M)$ from an empty cover, and the evolved program duly collapsed to infeasible covers (M of 50.7 digits at $E = 111$) — a textbook reward hack. With the E -cap restored in the fitness, the evolved move-set improved our best 168-digit blueprint by 0.9 digits at equal wall time; under warm-start conditions its Metropolis engine lost to our deterministic best-residue proposals. Verdict: harness patterns adopted, engine not — and the incident seeded the anti-reward-hack protocol that Chapter 7 builds on.

4.3 The scan engine, and three measured optimizations

The inner loop is fixed by the mathematics: for each k , sieve the residual complements by small primes, then base-2 Fermat-test the survivors, abandoning k at the first probable prime. A candidate costs about $1/\hat{p} \approx 17$ modular exponentiations regardless of survivor count, because k dies at its first PRP pass; winners are re-verified exhaustively with BPSW over *all* primes below Q before being reported. Three optimizations, each measured before deployment:

1. **Sieve depth.** Deeper sieving cuts tests per k like $1/\ln B_{\text{sieve}}$ but costs linearly; the sweep found $B_{\text{sieve}} = 10^6$ optimal on CPU (901 vs 816 candidates/s at the old $3 \cdot 10^6$ default), with the measured density invariant across depths, as it must be.
2. **Survivor-ordered testing.** Success probability at k is $(1 - \hat{p})^{a_k}$: hits concentrate exponentially in the low- a_k tail. Testing each block’s k in ascending survivor count, with stop-at-first-success, gave a measured $1.83\times$ conditional discovery gain at campaign scale ($12\times$ on

a high-density toy; 53% of toy hits in the bottom survivor-decile). The survivor count is a sufficient statistic for everything the sieve knows, a fact that returns with force in Chapter 7.

3. **Selective testing.** Since candidates are cheap to generate and expensive to test, skip the fullest ones entirely: with skip fraction $f = 0.92$, raw throughput rose $9.2\times$ ($1\,772 \rightarrow 14\,957$ candidates/s) while retaining 29.6% of the hit mass — computed *exactly* from the survivor histogram, so the density model stays unbiased under skipping — for a net $3.0\times$ discovery rate. The measured retained mass matched the histogram prediction to ± 0.4 percentage points at every f tried.

On GPU the same pipeline becomes a bit-matrix sieve feeding Montgomery-CIOS Fermat waves; porting optimizations 2–3 to the fleet was identified as the cheapest remaining nat in the programme.

4.4 Fleet discipline: compute that survives its container

Both pipelines ran on infrastructure that dies routinely: idle containers are reclaimed, cloud notebook sessions are culled, processes vanish mid-slice. The discipline that made multi-day campaigns possible on such substrate:

- **Slices as transactions.** The k -space is cut into slices (e.g. 262 144 values, ~ 2 minutes each) interleaved round-robin across catalog variants; a slice either completes and is checkpointed, or is re-run. Interleaving makes the global scan breadth-first in k , which also biases the first hit toward small k — exactly the threshold game’s preference.
- **Git as the checkpoint store.** Search logs with resume marks (`safe_k`, per-slice records) are *committed* before anything else happens. The GPU campaign survived roughly ten session culls and several container rollbacks with zero coverage loss; the CPU campaigns absorbed ~ 15 restarts the same way. Detached processes and even harness-tracked background tasks measurably did not survive (duty cycle 12%); timeboxed foreground chunks with committed checkpoints ran at $\sim 85\%$.
- **Verification at the edge.** Hits are re-verified on the worker before being reported, so a lost session cannot lose a record — harvest first, then die.

4.5 The certification standard

Every record passes five legs, uniform across pipelines:

1. **Reconstruction:** N is rebuilt by CRT from the published congruences and search index and compared byte-for-byte against the published decimal.
2. **Exclusion:** every prime $p < \mathfrak{g}$ is re-proven a non-summand from scratch — proper congruence or trial divisor, else a failed strong base-2 test — with the *count* of scanned primes asserted against an independently computed $\pi(\mathfrak{g})$.
3. **Partition:** $\mathfrak{g}$ is proven prime directly; $N - \mathfrak{g}$ ’s ECPP certificate is validated by PARI (`primecertisvalid`) and re-verified by an independent implementation (projective/affine elliptic-curve arithmetic with explicit gcd surfacing, exact $q > (n^{1/4} + 1)^2$ bound checks, chain closed at 2^{64} by deterministic Miller–Rabin), including the binding of the certificate’s top integer to $N - \mathfrak{g}$. CPU-pipeline records add APR-CL as an algorithmically distinct second proof and a PARI `ispseudoprime` full scan of every $p \leq \mathfrak{g}$ as a third code path.

Table 4.1: All certified bounded-game constructions of this thesis, in order of appearance. “Odd congr.” excludes parity; $|U|$ is the residual count below the cover’s target; the search index is k (or t) in $N = N_0 + kM$. Game column: T = improved the threshold record when found, R = improved a budget record, (T)/(R) = certified but superseded at merge time. Pipeline: C = CPU, G = GPU fleet, 1 = first generation (Chapter 2).

digits	$\mathbf{g}(N)$	game	odd congr.	digits M	$ U $	index k/t	pipe
1020	100 747	baseline	355	1020	0	0	1
237	109 621	height	102	232	610	299 948	1
199	105 667	T,R	89	193	—	876 554	1
199	110 917	T,R	88	191	729	299 581 384	C
197	107 719	T,(R)	88	191	752	951 928	C
195	100 297	(T)	88	191	706	70 186	C
193	102 337	T	86	185	686	27 994 707	C
186	109 357	T	83	177	709	82 846 821	C
179	101 149	T	80	170	728	758 850 385	C
150	104 527	T	68	140	827	62 147 038 260	G
200	112 249	(R)	89	195	765	56 770	C
199	119 419	R	88	190	806	145 605 823	C

4. **Manifests:** SHA-256 hashes over all artifacts.

5. **Audit:** an independent session re-ran the verifiers and mounted deliberate corruptions — a flipped digit in N , a deleted witness, a swapped certificate candidate, and a colluding expected-value edit — each of which the stack detected loudly.

The standard was not decorative. Twice in the programme’s history a check that skipped a leg returned a false PASS and the remaining legs caught it: a multi-line GP loop that silently parsed to a vacuous scan (caught by the prime-count assertion), and a thread-stack overflow that left a 2-byte junk “certificate” (caught by type-checking plus the independent re-verification). The audit also found three false-*reject* bugs and zero false-accept paths — the right asymmetry, inherited from one-sided witnesses: strong-test failures and gcd-surfacing arithmetic can wrongly reject, but not wrongly accept. No record that passed all five legs has ever been overturned. Verification costs minutes per record; Appendix A gives the exact commands.

4.6 The certified ladder

Table 4.1 is the full catalogue; the narrative below walks it. Parallel sessions occasionally produced overlapping rungs — the 195-digit record landed while a sibling session already held 186 digits — and the merged board always kept the best; superseded rungs are retained as certified constructions because their statistics validate the model and their covers seed later variants.

Two birds, one integer. The first thesis records were *dual-game*: a 199-digit N with $\mathbf{g} = 110\,917$, built to slip under the first-generation N_{199} (it does — by 0.05%, which is what happens when a hit arrives at 99.9% of the committed range, $3.0\times$ the selection-adjusted expectation; §3.8’s cautionary tale), and a 197-digit N with $\mathbf{g} = 107\,719$, found after $1.4 \cdot 10^8$ elements across 165 progressions at measured $\hat{E} = 18.2$ against predicted 18.20.

The threshold descent. A pure threshold cover needs only $Q = 100\,003$, not the dual game’s 105 668: 8% fewer targets, $|U|$ from 752 to ~ 706 , three times less search (§3.6). Stacked with survivor-ordered testing and the sieve optimum, a two-catalog campaign (1 590 progressions, $1.75 \cdot 10^8$ sieved elements, $\hat{E} = 17.4$ throughout) delivered a 195-digit record with $\mathbf{g} = 100\,297$ on a 4-core box in ~ 30 hours of scan — and, as the tail of the same catalogs, a second independent hit ($\mathbf{g} = 101\,611$ at a larger N), closing the campaign at 2 hits against a cumulative expectation of 2.1. Meanwhile the sibling CPU session ratcheted $193 \rightarrow 186 \rightarrow 179$ digits, each rung costing $\approx 2.5\times$ the previous search, exactly the frontier slope. The 179-digit rung was a single progression scanned to $t = 7.6 \cdot 10^8$ — it survived $\lambda \approx 2.5$ expected hits (an 8% draw) before landing, the depth-first design whose unbounded runway converts bad luck into digits rather than termination.

The GPU rung. At 150 digits the exponent is $E = 24.91$: about $e^{7.4} \approx 1\,600\times$ more search per hit than at 195. That rung was unreachable by the CPU pipeline and fell to the CUDA engine: the full round $[0, 7.5 \cdot 10^{10})$ was sharded across a rotating two-session fleet, scanned end to end ($\sim 2.1 \cdot 10^{12}$ Fermat tests), and produced exactly one hit at $k = 62\,147\,038\,260$ — 63.8% into the round, cumulative expectation 0.40, a central Poisson draw. The record:

```
8266689636142221488914017348682490933831701805981351031435344058125034
2747393905081255580217674034223624916359752668869952172293093032932050
5713336368
```

$N_{150} \approx 8.27 \cdot 10^{149}$, $\mathbf{g}(N_{150}) = 104\,527$: the smallest known even integer whose least Goldbach summand exceeds 100 000, 49 digits below the first-generation record. Evidence: all 9 977 prime offsets below 104 527 excluded (parity 1, congruence divisor 9 097, trial divisor 433, strong base-2 witness 446), 104 527 prime, complement ECPP-certified.

The budget climb. Under a hard ceiling the game is range-bound (§3.5): covers are re-tuned to leave k -room, catalogs supply the progressions, and every hit’s *first surviving offset* usually overshoots the cover target by thousands — free upside. The 199-digit budget record,

```
2461381565223403757902559202409729839707029814556797195332164743319183
6055009475885630448183435105080229622243237603782026136003397685621411
00015486692952424689982503380965035816275690874367307430972
```

has $\mathbf{g} = 119\,419$ from a $Q = 114\,000$ cover (88 odd congruences, $|U| = 806$, $t = 145\,605\,823$), bounding $R(10^{199}) \geq 119\,419$ (and a fortiori $R(10^{200})$). A parallel campaign at $Q = 107\,720$ produced a 200-digit N with $\mathbf{g} = 112\,249$ — found in the 16% of sieve blocks that overshoot the 10^{199} ceiling, a fair draw at measured $\hat{E} = 18.4$ against predicted 18.3, now dominated but kept: its statistics are part of the validation ledger.

4.7 Negative results, reported like records

Three campaigns ended without a hit, and one audit prevented a misallocated one; their numbers appear in the validation of §3.3 and their lessons in the design rules above.

- **Campaign 1** ($Q = 107\,720$, dual-game, 200 variants): retired strictly obsolete at 1 213 of 2 400 slices when the master merge brought stronger records; $3.18 \cdot 10^8$ elements, zero hits, survival probability 0.30 — unlucky, ordinary, and fully on-model per-slice.
- **Campaign 2** ($Q = 100\,003$, 400 variants capped at 175 digits): stopped at $5.6 \cdot 10^8$ elements, zero hits, $P \approx 0.22$. Post-mortem against the winning 179-digit single-progression design:

breadth-first variant pools cap the best possible outcome and can exhaust; depth-first ratchets dominate under fixed compute. Adopted.

- **Campaign 3** ($Q = 122\,000$, budget game): stopped early by coordination — a sibling fleet was already attacking the same game at higher Q — at 380 of 17 200 slices, hit-mass $\lambda = 0.077$, in-model throughout. Independent workers must scan *disjoint* variants; duplicated k -ranges are pure waste. A committed variant-ledger became protocol.
- **The blueprint audit.** An externally contributed 168-digit campaign blueprint was audited before adoption: the cover was valid, but its claimed E was optimistic by a nat (19.84 honest vs 20.85 claimed), its expected hit was 176.5 digits (not 168), and its shipped variants carried a ceiling that left $\sim 10^3 \times$ too few candidates to matter. Re-parameterized, it became Campaign 2. Moral: audit contributed specs with the same machinery as records — the cost model prices claims, not intentions.

4.8 Reproduction

Every construction in Table 4.1 is reconstructable as $N = N_0 + kM$ from its committed record file, and every witness is re-derivable by the standalone verifiers — nothing depends on the discovery path. Appendix A lists the one-command verification for each record and the SHA-256 manifests. The next chapter takes the same machinery to the game with no size limit at all, where the opponent is no longer the search budget but the primality certificate.

Chapter 5

Goldbach deserts are not prime gaps: the height game

A prime gap of length Q just below N forces $\mathfrak{g}(N) > Q$, so the height game could in principle free-ride on the prime-gap records. This chapter shows it should not: blocking only the prime offsets is strictly cheaper than blocking an interval, and the separation is now certified, not heuristic. Two constructions — $\mathfrak{g} = 1\,157\,341$ at 2480 digits and $\mathfrak{g} = 1\,134\,871$ at 2692 digits, from independent pipelines — exceed the longest prime gap ever certified with proven endpoints (1 113 106, between primes of 18 662 digits) using integers roughly seven times shorter than that gap’s own endpoints. The second construction additionally carries a fully certified local prime gap of just 8970: its Goldbach desert is 126.5 times longer than the prime-free interval it sits in, so the desert owes essentially nothing to an ordinary gap. At these heights the search is nearly free ($E \approx 6.6$) and the binding constraint becomes the primality certificate for the partition — making the height record an index of primality-proving technology rather than of number-theoretic difficulty — and we close with the one construction idea that would remove that wall too.

5.1 The transfer, and its one-way sign

Proposition 2.14 sends prime gaps to Goldbach deserts: if $[N - Q, N - 2]$ is prime-free then $\mathfrak{g}(N) > Q$ (given any partition at all). The converse direction is where the money is, and it fails constructively: a desert needs $N - q$ composite only for the $\pi(Q)$ primes $q < Q$, a set of density $1/\ln Q$ inside the interval the gap must clear entirely. Three certified exhibits measure the separation.

Exhibit 1: a desert that is a gap in disguise. The primorial $m_{43} = 13\,082\,761\,331\,670\,030$ of Remark 2.7 has $\mathfrak{g}(m_{43}) = 89$, and the nearest prime below it is $m_{43} - 89$ itself: this small desert coincides with an ordinary prime-free interval. At toy scale the two phenomena are entangled.

Exhibit 2: a desert 57 times its gap. The first-generation 237-digit record ($\mathfrak{g} = 109\,621$) already separates them: the committed verifier proves the interval $[N - 1926, N - 2]$ prime-free while $N - 1927$ is a probable prime (1927 is composite, so it is no summand). Ordinary gap information therefore certifies only $\mathfrak{g} > 1\,926$; the actual value is $57\times$ larger, entirely on the back of composite offsets that the desert may ignore.

Exhibit 3: the certified ratio. The 2692-digit construction below carries *both* numbers as certificates: desert 1 134 871, local prime gap 8970. Ratio: 126.5.

5.2 Benchmarks from the gap world

The longest prime gap known with *proven* endpoints has length 1 113 106 and sits between primes of 18 662 digits; the longest known with probable-prime endpoints has length 16 045 848 [14]. Two framing facts. First, certifying a gap is brutally expensive at scale: the endpoints need primality proofs, and every interior integer a compositeness witness — which is why the proven-endpoint record lags the PRP record by an order of magnitude. Second, gap *merit* (length divided by $\ln$ of the endpoint) measures how anomalous a gap is: the largest merit ever observed is ≈ 42 , and Cramér–Granville-type heuristics [31] cap plausible merits near a few hundred. Hiding a million-length desert inside a genuine prime gap at 200 digits would need merit $\approx 2\,180$ — fantasy. Deserts at practical sizes *must* come from covering primes, not intervals; this chapter’s constructions are that statement in certified form.

The challenge the height game set itself: beat 1 113 106 — a desert longer than any interval ever proven prime-free — with integers small enough to certify on a lab box. (The challenge benchmark was set at 1 113 137, marginally above the record gap; both constructions clear both numbers.)

5.3 The 2 480-digit record: $\mathbf{g} = 1\,157\,341$

The CPU pipeline’s height construction uses 759 odd congruence classes plus parity ($M \approx 10^{2474.6}$), leaving $|U| = 2\,844$ residual primes below the target $Q = 1\,113\,137$, and scans the progression $N = N_0 + tM$. At 2 480 digits the exponent is tiny — $p_1|U| \approx 7.7$, expected $\approx 2.2 \cdot 10^3$ progression elements, $\approx 1\,800$ *tested* elements under selective testing — and the hit arrived at $t = 14\,544$, the 443rd tested candidate. Certification is where the effort lives:

- every one of the 89 840 primes $p < \mathbf{g}$ proven a non-summand: 86 714 by covering divisor, 3 126 by failed strong probable-prime test, all re-derived by the standalone verifier;
- $\mathbf{g} = 1\,157\,341$ prime by trial division; the complement $N - \mathbf{g}$ (a 2 480-digit integer) proven prime by a 245-step ECPP certificate (≈ 1.7 hours to generate), re-verified from scratch by the independent pure-Python checker (7 036 s), *plus* APR-CL as an algorithmically distinct second proof (7 409 s);
- an independent PARI full scan of all 89 841 primes $p \leq \mathbf{g}$: zero violations (27.4 minutes).

Result: $\mathbf{g}(N) = 1\,157\,341$, exceeding the certified-gap record by 4% with an integer $7.5\times$ fewer digits than that gap’s endpoints. The desert record and the gap record are now formally decoupled: ours needed no interval at all.

5.4 The 2 692-digit mega-desert, and its certified local gap

The GPU-session pipeline independently built a second height construction — 813 odd congruences plus parity, 2 692 digits, found at $k = 14$ with measured $\hat{E} = 6.56$ against predicted 6.6 — with

$$\mathbf{g}(N) = 1\,134\,871,$$

proven by 88 239 offset witnesses (parity 1, congruence divisor 85 476, trial divisor 895, strong base-2 1 867) and an ECPP-certified complement, the 253-step certificate re-verified by the independent checker with its binding to $N - \mathbf{g}$ asserted.

What makes this construction scientifically distinctive is the *extra* pair of certificates: the nearest primes around N are pinned at $N - 7\,329$ and $N + 1\,641$ — each carrying its own validated ECPP certificate, with every interior integer deterministically composite — so the ordinary prime gap containing N has length exactly 8 970, fully certified. The desert-to-gap ratio,

$$\frac{\mathfrak{g}(N)}{\text{local prime gap}} = \frac{1\,134\,871}{8\,970} = 126.5,$$

is the thesis’s cleanest single number for the claim that Goldbach deserts are not prime gaps: 99.2% of this desert’s length lies *outside* any prime-free interval, bought by covering prime offsets alone. In digit terms the separation reads: a 2 692-digit integer carries a 1.13-million desert, while an interval of that length free of primes is only known (unproven!) around integers of $\sim 18\,000$ digits — the sparse-cover advantage is about $7\times$ in digits today and widens with the target.

5.5 The economics of height: certification is the wall

The cost model explains why the height game is played at thousands of digits. $E = |U| \cdot \text{boost} / \ln N$ *falls* as N grows — doubling the digits roughly halves the exponent at fixed cover strength — so unlike the digit-frugal bounded games, the height game optimises at whatever size makes the search trivial: $E \approx 6.6$ at 2 700 digits means hits arrive by the dozen. What does not get cheaper is proving the partition:

- ECPP generation and validation are minutes at 200 digits, ≈ 2 hours (plus 2 hours of independent re-verification, plus 2 hours of APR-CL) at 2 480–2 692 digits, and scale roughly quartically in the digit count;
- at $\sim 4\,000$ digits, days; far beyond that, the certificate — not the desert — is the record’s binding constraint with present tools.

The height record is therefore an index of primality-proving technology. Extrapolating with the cover economics: present ECPP practice supports $\mathfrak{g} \approx 2.5$ million (the next committed rung); distributed fastECPP-class certification ($\sim 50\,000$ -digit proofs) would support $\mathfrak{g} \approx 5 \cdot 10^7$; a certification breakthrough to 10^6 -digit general-form proofs would carry $\mathfrak{g}$ toward 10^9 . The desert machinery itself is already there: even the absolute-gap milestone — beating the 16 045 848 PRP-endpoint gap, i.e. covering $\pi(1.6 \cdot 10^7) \approx 1.03$ million prime offsets with a $\sim 20\,000$ -digit modulus — is a throughput problem (~ 0.2 s per modular exponentiation at that width; a distributed campaign 10^2 – $10^3\times$ one lab box), not a conceptual one. The certificate is the only wall.

5.6 Removing the wall: pick the provable prime first

Every construction in this thesis searches over N and pays ECPP for whichever complement $N - \mathfrak{g}$ falls out — a random 2 700-digit prime with no exploitable structure. Invert the design: *fix the complement first*, inside a family whose primality is cheap to certify, and search the other side.

Concretely, take Proth-form candidates $P = j \cdot 2^n + 1$ with $j < 2^n$: Proth’s theorem [21] certifies such a P prime with a *single* modular exponentiation once a quadratic non-residue witness is found — minutes at 10^5 digits, where general-form ECPP is hopeless. For fixed n , the cover congruences $N \equiv a_r \pmod{r}$ with $N = P + s$ (s the intended least summand) translate to congruences on j , since $2^n \pmod{r}$ is computable; the search over j is again a CRT progression, now on the *certifiable* side. A hit hands over the partition $N = s + P$ with P ’s certificate essentially free, deleting

the height game's only wall (Pocklington-style variants [20] offer the same trade with partially factored $P - 1$). The open questions are the density of Proth primes inside the cover-constrained j -progression and whether the n -side and s -side constraints can be tuned jointly; nobody has run the construction, and we flag it as the single most valuable open trick in this problem family — it would let the $\mathfrak{g}$ -ladder run to Proth-search scale (10^5 – 10^6 digits, $\mathfrak{g} \sim 10^8$ and beyond) with today's tools. Chapter 8 files it among the open problems with its neighbours.

Chapter 6

The limits of compression: why sub-100 digits is out of reach

The calibrated truth for $T(100\,000)$ sits near 10^{53} ; the certified record is 150 digits. This chapter measures how much of that logarithmic factor of three is closable, and proves — in the measured, adversarially tested sense this thesis uses — that the answer is: not much. Four attacks on the size of N are developed to their ceilings. Joint optimization of the CRT representative buys exactly its search entropy (24.3 of a provable 24.9 digits on the record cover) and cannot compete with progression scanning. Set-valued conditioning buys 1–2.8 nats, its extreme-value promise collapsing to $H^{0.35}$, and the classic cover emerges as the unique freely-enumerable design. Oversized covers would make sub-100 trivial ($E = 10.2$) but finding their small representatives is modular subset-sum with per-position choices at density ≈ 1.026 — the hardest known regime, with the best generic attacks at 2^{938} here — and the polynomial-time CRT-decoding escape is closed twice: by a coupling argument (with a retraction recorded of our own earlier claim to the contrary) and by experiment, practical lattice reduction failing to penetrate the Johnson radius even on solution-abundant instances. The only viable route to sub-100 digits is brute force at $E = 44.1$: about $9 \cdot 10^5$ GPU-years. The chapter closes with the walls of all three games and what protects each.

6.1 The question, and the two axes of attack

The threshold game is a compression problem: pack “every prime below 10^5 fails as a summand” into as few digits as possible. The digit-budget identity (§3.4) says a construction spends $\ln N = \ln M + \ln k$; the frontier (Table 3.2) prices the paradigm’s trade between them. Beating the frontier means one of two things: making the *modulus* side cheaper (better covers, or moduli whose representative is smaller than it deserves to be), or making the *search* side cheaper (conditioning that raises the success density). This chapter runs both axes to their measured ends; Chapter 7 then closes the third axis — replacing congruences altogether — with a theorem.

6.2 Attack 1: small representatives by construction

A CRT system’s least representative x is what the record pays for; the modulus M is what covers. Could a system with M far *above* the budget have x far below it — oversized coverage smuggled under the ceiling?

No free small representatives. Over 2000 random top-residue reassignments of the 197-digit record’s cover, the deciles of x/M sit at (0.10, 0.25, 0.50, 0.75, 0.90) — exactly uniform, with the sample minimum at the 1/samples order a uniform draw predicts. Residue choice carries no structural bias toward small representatives; only *search entropy* — enumerating many assignments and keeping the smallest — buys them.

Buying with entropy, at the exact exchange rate. The tool is a multiple-choice modular meet-in-the-middle in the style of Wagner’s k -tree [27]: split a block of moduli into 2^d lists, enumerate the top- c residue classes of each, and merge with shrinking centered windows until the representative lands in a target interval. The entropy accounting predicts 2^d lists of size L cancel about $(d+1)\log_{10} L$ digits; measured on the record cover: a 2-list split cancelled 8.3 digits at a cost of +8 residuals (46 solutions found where entropy predicted ≈ 43); a 4-level tree cancelled 21.3 digits (52 vs ≈ 50); the deepest run cancelled 24.3 digits against its 24.9-digit entropy ceiling — within half a digit, at ~ 25 minutes and 8 GB. Three certified by-products (committed and re-verified end to end) put oversized moduli under the 10^{197} budget:

Table 6.1: Joint representative/coverage constructions: modulus far above the budget, representative under it. The 211-digit system is a strictly better cover than the record’s own ($|U|$ 738 vs 752).

system	odd moduli	digits M	digits x	$ U $	E at 10^{197}
m211_x196	95	211	196	738	18.09
m219_x197	98	219	< 197	784	19.33
m221_x197	98	221	< 197	782	19.32

Why this does not improve any record. Extend the objective to expected hits, $\Phi = E - \ln K(x, M; B)$, where $K = \max(0, \lfloor (B-1-x)/M \rfloor + 1)$ counts progression elements under the budget. A sieved progression scan generates $\sim 1\,850$ candidates per second at $\Phi = 18.20 - \ln(1\,371\,832) = 4.07$; k -list construction generates 0.1–100 candidates per second of $K = 1$ each — candidates 10^1 – $10^4\times$ more expensive with E no lower, $\Phi \geq 18$. Joint optimization is the right tool exactly when the modulus *must* exceed the budget, and there it caps at the search entropy. The cap is brutal at scale: compressing the 2480-digit height cover to 197 digits would need $\sim 10^{2495}$ enumerated states — matching the uniform-draw estimate $10^{197-2480}$ head-on. Attack 1 is closed by measurement: the exchange rate is exactly entropy, and entropy is unaffordable.

6.3 Attack 2: set-valued conditioning, and the enumerability theorem

The classic cover pins $N \bmod r$ to *one* residue, paying $\ln r$ nats; conditioning it into a *set* A_r of k_r residues pays only $\ln(r/k_r)$. A Gaussian extreme-value argument promises coverage gains of order $\sqrt{2H \sum_r P_r}$ for H nats spread thinly — orders of magnitude, if attainable. It is not, and the reasons are structural.

The exact model (survival as a product over independent $N \bmod r$, reproducing the validated $E = |U| \cdot \text{boost} / \ln N$ to 0.3% at $k_r \equiv 1$) gives, optimised by Lagrangian sweep over every prime modulus below 15 000:

The promised $\sqrt{H}$ law measures as $H^{0.35}$, for two reasons: *overlap* (the nominal kill mass $\sum_r P_r \approx 25\,000$ chases only $\pi(Q) = 9\,591$ targets, so most extreme-value gain re-covers covered

Table 6.2: Set-valued conditioning at $Q = 100\,003$: the entire achievable gain.

D (digits)	classic E	set-valued E	gain (nats)
100	43.84	41.83	2.0
120	34.25	32.67	1.6
130	31.62	29.23	2.4
140	29.36	26.53	2.8
150	25.55	24.47	1.1

primes — the product structure eats independent contributions) and *Poisson discreteness* (large-modulus classes hold 0 or 1 targets; there is no excess over the mean left to collect).

And even those nats are not real, because of *enumerability*. The set-valued optima condition ~ 200 moduli of product $\sim 10^{450}$: representatives below 10^{100} exist, but finding which residue combination lands there is precisely Attack 1’s problem again. Free enumeration requires the conditioned modulus $M_1 \leq B$, in which case the candidate pool is $\text{pool} = (B/M_1) \cdot \prod_r k_r$, and optimising under that constraint the multiplicity term is nearly free but *redundant*: at 140 digits it contributes $e^{12.4}$ of pool for 0.15 nats of E , while plain k -room already prices candidates at 0.05 nats of E per nat — net gain ≈ 0.2 nats. The structural conclusion deserves italics: *the classic cover is optimal not by luck but by construction — it is the unique design whose admissible set is an arithmetic progression, i.e. free to enumerate; every relaxation that buys coverage destroys enumerability, and every relaxation that preserves enumerability buys almost nothing.*

6.4 Attack 3: oversized covers, and the subset-sum barrier

Suppose we ignore enumerability and aim directly at the dream: covers whose mass far exceeds the budget. At $B = 10^{100}$, $Q = 10^5$:

- **Regime I** (modulus under the bound): the honest optimum is 44 moduli, $M = 10^{79}$, $|U| = 1\,060$, $E = 44.1$ — $1.4 \cdot 10^{19}$ candidates.
- **Regime II** (oversized): 420 moduli with 16 allowed classes per position give $|U| = 165$ and $E = 10.2$ — about 27 000 candidates. Sub-100 would be a lunch-break computation.

Regime II’s design space carries $\approx 2\,698$ nats of entropy against $\ln(M/B) \approx 2\,630$: about $e^{57} \approx 2^{82}$ valid (design, N) pairs *exist* below 10^{100} . But the density of designs whose representative lands under the bound is e^{-2630} , and finding one is exactly:

$$\text{minimise } \left| \sum_r c_r \bmod M \right| \text{ over per-position choices } c_r \in C_r, |C_r| = L, \text{ target window } [0, B)$$

— inhomogeneous modular subset-sum with choices, at density (design entropy)/ $\ln(M/B) \approx 1.026$. This is the worst possible regime: low-density instances ($d < 0.94$) fall to lattice reduction [25, 26], very high densities to birthday/ k -tree merging [27], and density ≈ 1 is where subset-sum’s presumed hardness concentrates. Concretely here: meet-in-the-middle 2^{1897} ; Wagner’s k -tree with the available list sizes 2^{938} ; both astronomically short of 2^{82} planted solutions in a 2^{3893} space.

The decoding escape, closed — with a retraction. One hope remains: treat the residues as a Chinese Remainder code and use Guruswami–Sahai–Sudan list decoding [23] (in Howgrave-Graham’s lattice formulation [24]) to *decode* a small representative directly. The verified radius:

agreement mass $A > \sqrt{L \ln B \cdot P}$ suffices, P the log-mass of the whole modulus pool (numerically confirmed against the exact finite- ℓ optimum to 0.4%). Version 1 of our analysis claimed an open feasibility window for this attack. **That claim was wrong and is retracted:** it compared the decoding radius at one agreement size against an existence bound optimised over a *different* size. Coupling the parameters closes the window at every pool size — the barrier in one breath: *decodable* $\Rightarrow$ *pool small* $\Rightarrow$ *subset entropy too small for a representative to exist; representative exists* $\Rightarrow$ *pool large* $\Rightarrow$ *radius far beyond the cover mass*. Re-derived independently (Legendre-transform design entropy, same-mass coupling):

Table 6.3: The decoding barrier: maximum cover mass for which representatives exist vs minimum mass the decoder needs, same pool and L . The window never opens; the best case is 28% short.

pool R	L	pool mass P	$A_{\max}$ (exists)	A_J (decodes)	ratio
1 000	1	956	339	469	0.723
60 000	1	59 816	570	3 711	0.154
60 000	16	59 816	1 305	14 845	0.088
10^6	16	998 483	1 697	60 651	0.028

The gap widens with pool size and with L ($A_J \sim \sqrt{LP}$ against a logarithmic existence bound). The retraction’s lesson is recorded for the methodology chapter: *feasibility windows built from two separately-optimised bounds are worthless; couple the parameters first*.

6.5 Attack 3, empirically: LLL versus the Johnson radius

Proofs bound algorithms that exist; experiments bound hope. Could practical lattice reduction beat the proven radius on our *structured, solution-abundant* instances? A controlled study at toy scale: 61 positions (odd primes < 300 , $\ln M = 276.3$), messages below $B = 10^{12}$, Johnson radius $A_J = \sqrt{\ln B \ln M} = 87.38$; planted instances revealing mass ρA_J of true residues; LLL (FLINT, $\delta = 0.99$) on the Howgrave-Graham lattice at degrees $\ell \in \{12, 20, 30, 45\}$; 390 decodes total, predeclared protocol.

Results. Recovery is 0% at $\rho \leq 0.90$, 60% at $\rho = 1.00$, 100% at $\rho \geq 1.10$; measured against *realized* agreement mass the transition is razor sharp: at $\ell = 30$ every decode below $1.0271 A_J$ failed and every one above $1.0278 A_J$ succeeded — threshold $\approx 1.028 A_J$, tightening to $\approx 1.02 A_J$ at $\ell = 45$, tracking the finite-dimension determinant bound

$$A_{\min}(\ell) = \min_z \frac{z(z+1) \ln M + \ell(\ell+1) \ln B}{2(\ell+1)z}$$

to about 1% below it (the usual Coppersmith-family grace) and converging to A_J *from above* like $1 + c/\ell$. Below threshold the failure is total: the reduced bases yield integer-rootless polynomials — not the planted message, and not one of the 10^2 – 10^7 high-agreement alternatives that the first-moment count says exist at $\rho = 0.7$ – 1.0 . Across all 390 decodes, every integer root ever extracted was the planted message itself. *Solution abundance buys nothing*. The Johnson radius behaves as a hard practical barrier for this lattice family, closing the last “maybe practice beats the proof” hope from the barrier analysis.

6.6 The sub-100 verdict, priced

With Attacks 1–3 closed, sub-100-digit $T(100\,000)$ has exactly one route: Regime-I brute force at $E = 44.1$. Two independent costings (this thesis’s, and the parallel realizable-frontier study, which lands at $E = 43.66$ — agreement to 0.4 nats) put it at $1.4 \cdot 10^{19}$ candidates $\approx 9 \cdot 10^5$ GPU-years at the record engine’s measured rate — roughly seven orders of magnitude beyond plausible compute, and the number to quote against anyone’s “surely a bigger machine...”. The full menu, cap-aware (these E include the k -room overhead, hence sit about a nat above the raw frontier of Table 3.2; the model reproduces the actual 150-digit campaign — two GPUs, six days — as its calibration row):

Table 6.4: The descent below 150 digits, priced. Wall-clock rows from the measured fleet band ($7.9 \cdot 10^5$ candidates/s end to end); multiply by ~ 1.6 for 80% confidence.

D	E	candidates	A100-years	fleet wall-clock
150	25.5	$10^{11.1}$	0.01	~ 2 days (done: the record)
140	28.1	$1.6 \cdot 10^{12}$	0.10	~ 13 days
135	29.7	$7.7 \cdot 10^{12}$	0.49	~ 47 days
130	31.2	$3.6 \cdot 10^{13}$	2.3	~ 5 months
125	32.9	$1.9 \cdot 10^{14}$	12	—
120	34.7	$1.2 \cdot 10^{15}$	73	—
100	44.1	$1.4 \cdot 10^{19}$	$8.9 \cdot 10^5$	—

So the fundable frontier is 130–140 digits; 120–125 is the institutional edge ($\sim 10^2$ GPU-years); below that the record stalls on physics unless mathematics moves. Not impossibility — about 10^{53} qualifying integers exist below 10^{100} (§3.7’s accounting at this scale: the unstructured exponent is ≈ 110 against 230 nats of room) — but constructive inaccessibility, protected on two independent axes: search entropy in Regime I, and density-1 subset-sum in Regime II. For completeness, the quantum row: Grover [29] halves the exponent, but with resource estimates in the Gidney–Ekerå mold [30] a fault-tolerant oracle call (reversible 660-bit modular exponentiations, no early exit) costs $\sim 10^5$ – 10^6 seconds against the classical $1.25 \cdot 10^{-6}$: the crossover sits at $E^* \approx 55$ for one machine, ≈ 41 for a thousand — at or above every target we care about, with both sides infeasible at the crossover. The walls of this chapter are classical *and* quantum walls at any constants that have been projected.

Two well-posed problems would collapse the verdict, and are of independent interest: *CRT list-recovery beyond the Johnson radius for structured, solution-abundant instances*, and *modular subset-sum with per-position choices at density ≈ 1 when 2^{82} solutions exist*. They are filed in Chapter 8.

6.7 The walls, per game

Assembling this chapter with the calibrated truths of §3.9 gives each game its forecast. The table uses $\ln C \approx 25$ for today’s fleet, ≈ 35 for a nation-scale scan, ≈ 45 as a generous classical ceiling.

The budget game is where algorithms matter most per unit compute (its truth is only $\sim 15\times$ away); the threshold game is where they matter least (its truth is entropy-armoured from two sides); and the height game has no mathematical wall at all, only an engineering one. One axis remains unexamined by everything so far: perhaps congruence divisors are simply the wrong *mechanism*, and some algebraic structure certifies compositeness wholesale without paying $\ln r$ per class. Closing

Table 6.5: THE WALL: predicted limits per game. “Truth” is the calibrated Granville–van de Lune–te Riele estimate; the middle columns are what the validated machinery reaches at stated compute; “protected by” names the obstruction that survives everything this thesis threw at it.

game	truth	today / heroic	likely future	protected by
$T(10^5)$, smallest N	$\sim 10^{53}$ (53 digits)	$\sim 140\text{d}$ (days), $\sim 130\text{d}$ (months) / 120–125d	stall at 120–135d; floor 38–49d	search entropy; density-1 subset-sum; the closure triple (Ch. 7)
$R(10^{200})$, largest $\mathfrak{g}$	$\sim 1.8\text{M}$	$\sim 170\text{k}$ / $\sim 250\text{k}$	$\sim 450\text{k}$	range entropy under the cap
Height, largest $\mathfrak{g}$	unbounded	$\sim 2.5\text{M}$ / $\sim 5 \cdot 10^7$ (fastECPP)	10^9+ ; un- bounded if two-sided con- struction lands	certification cost only

that axis — or being handed a revolution by failing to — is the next chapter.

Chapter 7

The Divisor-Paradigm Closure Theorem

Everything before this chapter certifies compositeness one way: a planted congruence divisor. Could some algebraic mechanism — norm forms, cyclotomic identities, exponent families, polynomial factorizations — certify many complements composite wholesale, escaping the $\ln r$ -per-class accounting that walls in Chapter 6? This chapter answers with a theorem and its falsification engine. We formalise construction schemes as candidate families with certificate maps, prove that certificates either plant their divisor at design time or must extract it per instance (which is factoring), and show — across a concrete grammar $\mathcal{L}$ of planted mechanisms — that congruence covering is the unique economical one: value-set ceilings kill univariate identities at any price, a first-moment cluster bound caps every shared-variable identity family at $O(1)$ certified offsets, isolation and orbit accounting close the window-dense frontier, and a supply-side purchase argument closes the cheapest surviving family $2.5\times$ over budget. The engine swept 13 661 enumerated and 41 adversarially evolved instances: zero passes — but the adversary’s first find was a $50\times$ accounting hole in our own referee, and fixing it sharpened the theorem’s economics. With mechanisms closed, what governs the records is measured too: the entire game exponent is an integrality gap (the fractional cover is perfect), and the greedy point survives exact optimization over windows up to half the cover. Three doors remain — rounding mathematics, throughput, and two named conditionals — and we state each at its maximal form.

Rigor marks, used throughout this chapter: [P] proved at this thesis’s level of rigor; [C] conjectural; [V] empirically validated input with the dataset stated. The theorem’s honest status is assembled from these marks in §7.9.

7.1 Five probes before the theorem

Five bounded probes, each with a predeclared termination gate, tested the folk candidates for a better mechanism. The benchmark: covering certifies $\sim 4\,400$ offsets per nat of design entropy at $r = 3$, falling to ~ 1 per nat at the endgame ($r \approx 457$; measured marginal 1.01 nats per offset averaged over the last six moduli).

- P1** *Prime-power and composite moduli.* Terminated, strictly dominated: the class $q \equiv a \pmod{p^2}$ is a subset of the class mod p already killed, at double the cost [V: $p = 7$ gives $227 \subset 1\,613$ kills]. Composite moduli reduce to their prime factors with only losses.
- P2** *Polynomial value-set identities.* $N = m^k - c$ makes $N - q = m^k - d^k$ algebraically composite for every offset $q = d^k - c$. Terminated, dominated — but the closest call, and the seed of

Lemma 1: the best quadratic family kills 113 offsets below 10^5 [V: $c = 398$], yet confining N to a k -th-power family surrenders $(1 - 1/k) \ln B$ nats, and *the value set of a degree- k form below Q has $O(Q^{1/k})$ members* — at most a few hundred killable offsets at any price.

P3 *Sierpiński–Riesel exponent families.* $N_n = j \cdot 2^n + c$: for any fixed candidate the induced constraint is one congruence class — ordinary covering. The exponent structure correlates residues *across* the family, a legitimate search-throughput trick, not an entropy mechanism. Terminated.

P4 *Norm forms and splitting conditions.* “ $N - q$ is a norm in K ” does not certify compositeness (primes with the right splitting are norms too), and any usable splitting condition on q is a congruence condition mod discriminant-related moduli — covering in Galois clothing. Terminated.

P5 *The hidden-bias loophole.* Could constructed N make residual complements composite more often than the boost predicts? The banked campaigns answer: across three covers and $1.2 \cdot 10^9$ scanned candidates, $|\hat{E} - E| \leq 0.1$ nats [V]. Any unexploited structure is worth ≤ 0.1 nats. Terminated.

The probes sharpen the requirement. A paradigm-breaking mechanism must *simultaneously* (1) certify $\gg \sqrt{Q}$ distinct prime offsets, (2) cost $o(\ln r)$ design entropy per offset at the margin, and (3) be invisible in banked density data — activating only on constructed N . No known structure does all three. The rest of the chapter turns that requirement into a theorem about a language.

7.2 Schemes, certificates, and the theorem

Fix game parameters: bound B (here 10^{200}), threshold Q (here up to $2 \cdot 10^5$), $\ln N \approx \ln B$.

Definition 7.1 (Construction scheme). A *construction scheme* is a pair (F, C) :

- a *candidate family* F : a parameter set $\Theta \subseteq \mathbb{Z}^m$ with polynomial-size description and an evaluation map $N : \Theta \rightarrow 2\mathbb{Z} \cap [1, B)$ computable in $\text{poly}(\log B)$; its *design cost* is $\text{cost}(F) = \ln B - \ln |N(\Theta)|$, the nats of search freedom surrendered by joining the family;
- a *certificate map* C : on input (θ, q) , q prime $< Q$, outputs $\perp$ or a divisor witness $D(\theta, q)$ with $1 < D < N(\theta) - q$ and $D \mid N(\theta) - q$, computable and verifiable in $\text{poly}(\log B)$.

C is *planted* when the divisibility holds as an algebraic identity over the family (checkable symbolically); *extracted* when C discovers the divisor per instance. The *killed set* is $K(\theta) = \{q : C(\theta, q) \neq \perp\}$, the *residual set* $U(\theta)$ its complement in the primes below Q , and the *effective exponent* $E_{\text{eff}}(F, C)$ is the infimum, over polynomial-time searches, of the log of expected work per hit divided by work per candidate. For the classic cover, $E_{\text{eff}} = E = |U| \cdot \text{boost} / \ln N$ [V: §3.3].

The *covering frontier* $E^*(Q, B)$ is the minimum E over congruence covers at the boost-versus-kills equilibrium of §3.5 (e.g. $E^*(122\,000, 10^{199}) = 20.83$; $E^*(200\,000, \sim 10^{200}) \approx 37.7$).

Theorem 7.2 (Divisor-Paradigm Closure — target statement [C], proved on the grammar [P]). *Let (F, C) be a construction scheme whose certificate map is planted and expressible in the schema language $\mathcal{L}$ of §7.3. Assume:*

- (H1) *Hardy–Littlewood-type independence for shifted primes in families: conditional on all divisibility data by primes $< Q$ (and the sieve depth), the primality indicators of distinct residual complements are independent with the Mertens-adjusted rates;*
- (H2) *no compositeness test with constant soundness on random rough inputs runs in $o(\text{one modular exponentiation})$.*

Then $E_{\text{eff}}(F, C) \geq E^*(Q, B) - o(1)$. Moreover, extracted certificates on any window-dense family of targets are factoring-equivalent, and certificate-free detection is already employed at its known-optimal cost (one Fermat exponentiation) by the standard engine.

Informally: *within $\mathcal{L}$, nothing beats covering; outside planted- $\mathcal{L}$, a scheme must either factor or test, and both sit at known cost floors.* Before the lemmas, a sanity audit of everything outside the formalism, from the million-desert study (§3.7): perfect powers cannot help (consecutive squares near 10^{200} are $\sim 10^{100}$ apart, so the window contains none); hiding the desert in a genuine prime gap needs merit ≈ 2180 against ≈ 42 ever observed (§5.2); composite moduli are P1. The formalism is aimed at the only survivors: algebraic identities and per-instance discovery.

7.3 The schema language $\mathcal{L}$

The falsifiable core of the theorem is a concrete grammar over certificate mechanisms; each compiles to an offset enumerator, a symbolic divisor witness, and a design-cost term:

$\mathcal{L} ::= \text{Congruence}(d, a)$	$N \equiv a \pmod{d}$
$\text{PolyImage}(f \in \mathbb{Z}[m], \deg \leq 4)$	$N - q = A(m) \cdot B(m)$ identically
$\text{ExpFamily}(N = j \cdot a^n + c)$	per-offset multiplicative-order conditions
$\text{MultiPoly}(A, B \in \mathbb{Z}[u_1..u_v], \deg \leq 2, v \leq 3)$	$N - q = A(u) \cdot B(u)$
$\text{NormForm}(K, \text{disc} \leq 10^4)$	$N - q = \text{Norm}_K(\alpha)$, exhibited factorization
$\text{Cyclotomic}(\Phi_n, n \leq 200)$	cyclotomic/Aurifeuillian splittings
$\text{Compose}(\mathcal{L}, \mathcal{L})$	depth ≤ 2 .

Excluded by construction: “certificates” that need a primality decision to define membership (circular), and per-instance divisor discovery — the extraction horn, handled by the theorem’s second clause rather than by the grammar. Grammar extensions are a human act with a hardening rule (§7.7): any extension to $v \geq 4$ shared variables must ship with the dense-frontier analysis of Lemma 1(c) re-proved first.

7.4 Lemma 1: entropy accounting for planted certificates

Statement. For planted (F, C) , the amortised design cost per certified offset is at least $\ln(1/\delta_{\text{eff}}) - O(\ln \ln B)$, where δ_{eff} is the density of integers in a generic window of width Q at height B for which the scheme can exhibit its witness with polynomial-range parameters; and for every family in $\mathcal{L}$ this is no cheaper than covering. The proof is a case sweep, and the cases are the chapter’s real content.

(a) Congruence classes [P]. $\delta_{\text{eff}} = 1/d$, cost $\ln d$: the covering accounting itself, with composite d dominated by its prime factors (P1).

(b) Univariate identities [P]. A degree- k identity's certified offsets lie in the value set of a degree- k form below Q : at most $O(Q^{1/k})$ of them, *at any price*. At $Q = 122\,000$: 349 ($k=2$), 50 ($k=3$), 19 ($k=4$), 7 ($k=6$); and Φ_n has degree $\varphi(n) \geq 2$ for $n \geq 3$, so *Cyclotomic* sits inside the same ceiling. Both families are closed *price-free*: the engine's pass bar (below) demands $> 3\sqrt{Q} \approx 1\,048$ distinct offsets, and $349 < 1\,048$.

(c) Multivariate identities — the hard case. Here the representable set can be window-dense (products of two binary quadratic forms reach $\sim 9\,300$ of the 17 983 prime offsets at $Q = 200\,000$ [V]), so existence arguments must be replaced by *access* arguments. Write the factors' degrees $d_1, d_2 \geq 2$ (degree-1 factors are sieve-equivalent), $d = d_1 + d_2$, v shared variables, and $\alpha = \min(1, v/d)$ the value-set density exponent [V: measured $\hat{\alpha} = 0.761$ vs 0.750 predicted for $(2, 2, 3)$].

Cluster ceiling, $v < d$ [P modulo standard equidistribution of form values]. Distinct values of $G = A \cdot B$ below X number $\sim X^\alpha$; a first-moment count of k -clusters inside width- Q windows gives $X^\alpha (X^{\alpha-1} Q)^{k-1}$, so clusters exist only up to

$$k_{\max}(\alpha) = 1 + \left\lfloor \frac{\alpha \ln X}{(1 - \alpha) \ln X - \ln Q} \right\rfloor,$$

finite iff $\alpha < 1 - \ln Q / \ln X$ (threshold $\alpha^* = 0.9735$ at $X = 10^{200}$, $Q = 2 \cdot 10^5$). Every $v < d$ combination has $\alpha \leq (d-1)/d \leq 0.9 < \alpha^*$:

Table 7.1: The cluster ceiling: maximum certifiable offsets per candidate for shared-variable identity families (d_1, d_2, v) at $X = 10^{200}$, $Q = 2 \cdot 10^5$. A scheme needs $\sim 18\,000$.

combo	α	$k_{\max}$	combo	α	$k_{\max}$
(2, 2, 2)	0.500	2	(2, 2, 3)	0.750	4
(2, 3, 3)	0.600	2	(2, 3, 4)	0.800	5
(2, 4, 4), (3, 3, 4)	0.667	3	(2, 4, 5), (3, 3, 5)	0.833	6
(3, 4, 5)	0.714	3	any $v < d$	≤ 0.9	≤ 13

For the flagship case $(2, 2, 2)$ this is existence, not hardness: at $X = 10^{200}$ the expected number of 3-clusters is $Q^2 X^{-1/2} \approx 10^{-89}$, so a same-variable quartic-product scheme certifies *at most two* offsets per candidate with high probability over any construction ensemble — versus 18 000 needed. $\delta_{\text{eff}} = \Theta(X^{-1/2})$ is an existence statement, with the measured enumeration wall (work per certified in-window offset scaling as height $^\theta$, θ measured 0.50–0.74 against the theoretical 1/2 [V]; windows at height 10^{10} already contain zero enumerable hits at width 10^4) as its algorithmic shadow — $\geq 10^{100}$ operations per offset extrapolated to $B = 10^{200}$.

The dense frontier, $v \geq d$ [P at draft rigor]. When the value set is window-dense, two elementary propositions close access anyway. *Isolation*: around a solution u_0 with $G(u_0)$ in the window, any lattice displacement δ keeping $G(u_0 + \delta)$ within a width- $2Q$ window must satisfy $|\nabla G \cdot \delta| \lesssim Q$ and $|\delta^\top H \delta| \lesssim Q$; the solution slab has principal extents $Q X^{-(d-1)/d}$ and $Q^{1/2} X^{-(d-2)/(2d)}$ — at $d = 4$, $X = 10^{200}$: 10^{-145} and 10^{-47} . Every extent is below $\frac{1}{2}$, so the only lattice point is $\delta = 0$: *window solutions are pairwise isolated; there is no local navigation between certified offsets — dense existence is non-constructive by geometry, not by hardness*. *Orbit accounting*: an infinite group Γ of structured moves either preserves G (one offset per orbit), or preserves a factor — so every reachable value carries the planted divisor $a_0 = A(u_0)$: enumerable a_0 is a congruence certificate (covering economics), balanced a_0 is back in the isolation sliver — or mixes generators of both automorphism groups, whose coarse log-lattice of reachable factor pairs holds $\sim (\log X)^2 \approx 2 \cdot 10^5$ points against a target of relative width 10^{-195} : expected hits 10^{-190} . Any remaining access route

must solve $G(u) = m$ for *prescribed* m — but a product-form representation of m is a nontrivial factorization of m , so a prescribed-value solver factors the scheme’s own rough complements: the extraction horn, not a new hypothesis. Disjoint-variable products reduce directly to covering (the free small factor is the certificate; NormForm reduces to MultiPoly through exactly this route).

ExpFamily: closed by supply, not price [P, quantitative]. Order conditions are the cheapest certificates in the grammar: s divides $j \cdot a^n + c - q$ along an arithmetic progression of n when the discrete logarithm of $(q - c)j^{-1}$ lands in $\langle a \rangle \bmod s$, at design cost $\ln \text{ord}_s(a)$ — as low as $\ln 3 = 1.10$ nats. But $s \mid a^{\text{ord}} - 1$ forces $a^{\text{ord}} > s$, so cheap certificates need small orders in large s , and such s supply only $\sim \pi(Q) \text{ord}/s$ eligible offsets each [e.g. $s = 2801$, order 5: about 20]. Solving the resulting purchase problem exactly — sort all $s < 10^5$ by price, buy cheapest first until the pass bar of 1048 offsets — costs 1151 nats for every base $a \in \{2, 3, 5, 7\}$, against the entire $\ln B = 460$ -nat budget; the best *budget-feasible* design certifies 419 offsets. The strongest non-covering family in the grammar is $2.5\times$ over budget at the bar, ceiling 419 against 18000.

Compose: empty at scale [P]. Two nontrivial family constraints on the same N are contradictory: $N = m^2 - c_1 = m'^2 - c_2$ forces $(m - m')(m + m') = c_1 - c_2$, so $m \leq |c_1 - c_2|$, while $N \sim 10^{200}$ needs $m \sim 10^{100}$ [V: ordinary shift pairs have $O(1)$ solutions, largest $m = 1073$ in the swept instances]. The only surviving composition is family-plus-congruence — a capped family riding on ordinary covering, still under the bar. This explains structurally the additivity the engine measured (no super-additive interaction in any composed sample).

The economics, restated. The generative tier (§7.7) forced a sharpening: with every mechanism charged for *all* offsets it claims and capped by the budget, a bulk mechanism must deliver offsets at $< \ln B / (3\sqrt{Q}) \approx 0.44$ nats each — and the honest comparison is not covering’s *marginal* endgame (1.01 nats/offset) but its *average*: 437 nats buy 10607 certified offsets, 0.041 nats each. Covering wins not at the margin but on average, because $\ln r$ nats buy $\sim \pi(Q)/(r-1)$ offsets at once, whereas every known algebraic mechanism buys $O(1)$ offsets per unit of entropy.

7.5 Lemma 2: correlation exhaustion

Proposition 7.3 (Shared divisors are small [P]). *Let N be even, $q_1 \neq q_2$ odd primes $< Q$, and $d > 1$ with $d \mid N - q_1$ and $d \mid N - q_2$. Then $d \mid q_1 - q_2$, so every prime factor of d is $< Q$.*

Proof. Subtract. □

Consequence: every divisor event coupling two complements is measurable in the σ -algebra generated by small-prime divisibility — which is *exactly* the information the cover fixes (classes) and the sieve harvests (k -residues), and the engine consumes all of it: any “ k -side” congruence trick selects a sub-progression the sieve already identifies, and the survivor count is a sufficient statistic (§4.3). Under H1, the residual lottery then factorizes, E_{eff} decomposes as design cost plus $\sum_q p_q$, and that is the covering objective [P conditional on H1]. Empirical support for H1 at our scales is P5’s $|\hat{E} - E| \leq 0.1$ over $1.2 \cdot 10^9$ candidates [V]; the corollary worth stating is that *luck-shaping* — choosing N to make residual events favourably dependent — has no purchase beyond 0.1 nats, because dependence requires shared small divisors, which is covering by definition.

7.6 Lemma 3: the detector floor, posed as an open problem

The last escape is orthogonal to certificates: make *testing* cheaper than a modular exponentiation.

Open Problem 7.4 (H2: sub-modexp compositeness detection on rough inputs). *Let n be uniformly random among integers in $[X, 2X]$ with no prime factor below $(\log X)^{O(1)}$, and let $M(b)$ be the bit cost of b -bit multiplication. Does a randomized algorithm exist that decides compositeness of n with soundness $\geq 2/3$ (one-sided; “don’t know” allowed) in time $o(M(\log X) \log X)$ — asymptotically cheaper than one modular exponentiation? Variants: (i) constant recall r at cost $c \cdot \text{modexp}$ with $c/r < 1$; (ii) amortized: certify a constant fraction of the composites among $n_1, \dots, n_k$ in $o(k)$ modexps total, beyond the small-divisor mass reachable by product trees [28]; (iii) with polynomial advice depending on X only.*

Known art sits firmly at the floor: trial division and batch gcd have recall zero on rough inputs by construction; Jacobi symbols equidistribute (soundness $o(1)$); Fermat, Miller–Rabin, and Frobenius tests all cost $\Theta(M \log X)$; and no lower bound is known in any realistic model — even a conditional one in a restricted arithmetic model would be new. The problem has a live industrial consumer: the record engine spends $\geq 95\%$ of its time on exactly this primitive (~ 17 tests per candidate, $e^{20.8} - e^{37.7}$ candidates per campaign), as do the large distributed prime hunts; a detector at cost c and recall r multiplies throughput by $1/(c + (1 - r))$, and record difficulty is $\log(\text{compute})$. A positive answer breaks the cost floor of a whole genre of searches — which is the more interesting outcome, and why the problem is posed rather than assumed.

7.7 The falsification engine

A theorem whose proof is a case sweep earns its keep by inviting attack. The engine’s contract: a schema instance *passes* — i.e. threatens the theorem — iff simultaneously (1) it certifies $\geq 3\sqrt{Q}$ distinct prime offsets below Q (clears the univariate ceiling), with factor nontriviality spot-checked; (2) its cost per offset, charged for *every* offset claimed and budget-capped, beats the covering benchmark; (3) its predicted effect would be invisible in the banked density data (P5-consistency). Any pass is audited by hand before it counts. The scorer re-derives every number from primitives — deterministic sieve, the pinned residual set, symbolic factorization — and *proposals are data, never code*; fitness is capped, monotone in verified counts only, with no reward for prose, novelty, or runtime.

Tier 0 (calibration). The engine reproduces every P1–P4 hand number exactly, and surfaced two refinements worth keeping: the covering endgame is ≈ 1.0 nats/offset at $Q = 122\,000$ (an earlier 2.5-nat figure was context-bound), and the binding constraint crosses over from budget to equilibrium between $Q = 122\,000$ and $Q = 200\,000$ — both now part of §3.5.

The exhaustive tier. 13 661 schema instances across the full grammar: **zero passes**. Verdicts: 400 congruence-composite instances scored BASELINE (inside the paradigm — the paradigm cannot escape itself); 12 579 PolyImage rejected on the value-set ceiling (best instance remains $k=2$, $c=398$, 113 offsets); 36 ExpFamily, 324 + 4 MultiPoly, 60 NormForm, 198 Cyclotomic, 60 Compose rejected per the lemmas, with all composition overlaps within 3σ Poisson of additive. One methodological note kept for honesty: the first run produced 16 spurious passes — congruence-containing schemas leaking through the escape filter with fresh-universe kill counts. The fix was semantic (baseline family detection), not numeric; zero real passes before and after. Near-miss board: ExpFamily at

3.1–4.2 nats per residual kill — three times the covering endgame, exactly as P3 predicted — then nothing below 10.

The generative tier, and the referee it broke. The adversarial arm ran on a hosted evolutionary search service (AlphaEvolve), under a threat model written in advance: evolved candidates are *programs*, so their only contract is to emit schema JSON; they run sandboxed, never compute kills, never report fitness; grading happens in a fresh interpreter. Within about ten candidates the search drove apparent fitness $0.27 \rightarrow 0.46$ — not by finding mathematics, but by exploiting a hole in our scorer: per-offset costs were billed only against the 867 residuals of the benchmark cover, so an ExpFamily design could advertise 10 416 offsets costing 22 900 nats against a 460-nat budget and be credited anyway — a $50\times$ overdraft. The audit protocol caught it before any claim was made (a pass requires hand audit by construction); the fix — charge every claimed offset, cap coverage at $\ln B/\text{cost}$ — re-scored the champion to 0.046 and *strengthened* the theorem, yielding the budget constraint, the 0.041-nat average bar, and the purchase argument that closes ExpFamily quantitatively (§7.4). The run then converged: 41 candidates, four families, zero passes, best corrected fitness 0.0919 — the same near-miss family the exhaustive tier had already ranked first, tuned to the edge of what order conditions allow, stalled $53\times$ short of the bar.

Table 7.2: Generative-tier empirical ceilings per family under corrected, budget-aware accounting. “eff. offsets” is the budget-capped certified count; the pass bar is $> 1\,048$ offsets at < 0.44 nats/offset, against covering’s average 0.041.

family	best fitness	eff. offsets	nats/offset	vs the bar
ExpFamily	0.092	209	1.10–2.20	$53\times$ too expensive
Compose (2-deep)	0.076	695	8.86	$215\times$
Cyclotomic	0.039	348	8.71	$211\times$
PolyImage	0.001	56	46.1	$1118\times$

Neither structural fact in Table 7.2 is a search artifact: no family clears both filter halves at once (coverage is bought only at 8–9 nats per offset; cheapness only at $O(1)$ offsets per condition), and the ordering matches Lemma 1’s case analysis exactly. The methodological finding generalises beyond this thesis: *adversarial search against a formal claim is most valuable as a referee-hardening device, and should be reported that way, not as a failed refutation* — the designed behaviour of an audit protocol in which candidates emit data, graders run fresh, and no pass counts without human eyes.

7.8 Door 1: the rounding gap

With mechanisms closed, the covering objective itself is the last mathematical lever: how far from optimal are the covers? The answer reframes the whole game.

The gap is everything. The set-cover LP relaxation of the benchmark instance ($Q = 122\,000$; moduli fixed at the equilibrium set, the 88 odd primes 3..457; one residue class to choose per modulus) reports *fractional coverage* 100.00% — the classes carry $\sum_r \pi(Q)/(r-1) \approx 1.86 \pi(Q)$ of kill mass, and fractional routing covers every prime, at both $Q = 122\,000$ and $Q = 200\,000$. The LP certifies nothing about integral assignments; what it proves is a framing: **the entire game exponent is an integrality/rounding gap**. Naive independent rounding leaves $e^{-1.86} \approx 15.6\%$

of primes uncovered; adaptive greedy leaves 7.56%; the fractional optimum leaves 0. Every absolute percent of residual removed is ≈ 2.7 nats $\approx 15\times$ less compute.

The landscape, measured. Over the full e^{436} -point assignment ensemble, random assignments have $|U| \sim (\mu = 1\,377, \sigma = 21.0)$; the measured σ sits *below* the independence prediction (34.8) — negative correlations thin the tail. A first-moment (annealed) threshold lands at $u^* \approx 758$; greedy-plus-descent achieves 867, already $z = 24.3$ of the 29.5 available sigmas into the tail. Diagnostics: 19 of 88 moduli are choice-tight; random-start coordinate ascent lands 90–110 primes *worse* than greedy (adaptive construction is worth ≈ 2.4 nats over naive local search); and pairwise class agreement between independent solutions is 2.4% — chance level, a fully degenerate landscape with no backbone for crossover methods to graft.

Exactness, thrown at it. The greedy point survived everything: 41 exact CP-SAT [34] windows of 7–15 moduli (warm-started, incumbent-floored) — never beaten, always matched, with 21 of 22 computed LP window bounds *certifying* window-optimality (the one uncertified window contains the smallest moduli, the global LP’s fractional looseness in miniature); exact re-solves of the 30 and then 44 *largest* moduli simultaneously — half the cover in one subproblem — unchanged at 867; and belief-propagation-guided decimation (validated on a toy to the exact optimum) landing at 874, which its own local search cannot improve — a third mutually-isolated locally-exact basin (867/874/968), while the naive Bethe free energy slides toward the fractional fantasy at low temperature, sharing the LP’s blindness to integrality (a survey-propagation treatment [32] is the remaining theory item). Running verdict: 867 is at or very near the *quenched* optimum for this moduli set, the annealed 758 likely reflecting an annealed-versus-quenched (clustering) gap rather than reachable headroom; the honest Door-1 prize is 0–2.6 nats.

The open programme. What has *not* been tried is a different ensemble: the covering constructions of Erdős–Rankin and FGKMT [9, 11, 12, 13] are precisely a technology for rounding fractional covers of primes with quantified loss — layered smooth-kill middle bands, second-moment control, a hypergraph-matching endgame (cf. Moser–Tardos-style resampling [33] for the algorithmic version). Their loss is $o(1)$ in an asymptotic regime that is not ours (their moduli reach $y/2$; ours cap at 457), which is exactly why this is a thesis-grade open question rather than a lookup: *determine the achievable residual fraction at $\pi(Q) \approx 10^4$ targets under budget $\theta(457)$* . Closing even half the greedy-to-fractional gap is $e^{10} \approx 22\,000\times$ compute — the difference between a volunteer network and a GPU-month for the 200 000 rung.

7.9 Three doors, and the verdict

Every attack generated in this programme reduces to one of three irreducible doors; the closure argument is that there is no fourth. (i) Divisor information is fully harvested: k -side congruence tricks select sub-progressions the sieve already identifies. (ii) Correlation requires a shared divisor, hence is covering (Proposition 7.3), and P5 bounds everything residual at 0.1 nats. (iii) A compositeness certificate is planted (then Lemma 1’s accounting applies) or extracted (then it factors); and certificate-free detection already runs at its conjectured floor (H2).

Door 1 — rounding mathematics. Prize 0–2.6 nats on current evidence, with the FGKMT-transfer ensemble as the genuine unknown; every nat multiplies every future rung. Cheap, and the covering monopoly theorem is what makes investing in it safe.

Door 2 — throughput. The search is progress-free, stateless per ticket, with millisecond verification: the best-shaped volunteer workload there is, and the model on which the neighbouring record-sports (GIMPS, PrimeGrid — themselves covering problems in the Sierpiński–Riesel line) built everything. At the historical peak of volunteer computing ($\sim 2 \times 10^5$ of our fleets), the $Q = 200\,000$ rung falls in about two days; at 1% of it, months. The capital-intensive twin is a Montgomery-modexp ASIC rack ($\sim e^6$ over the fleet). The quantum door is closed at any projected constants: Grover’s halved exponent meets a $\sim 10^{11}$ -fold oracle slowdown, crossover $E^* \approx 41\text{--}55$, at or above every target of interest.

Door 3 — the paradigm. This chapter. Status: **proved on the swept grammar** [P at draft rigor] — every family closed by a proof of its own kind (value-set ceilings, cluster bound, isolation plus orbit accounting, purchase argument, empty intersection) — **conjectural at the boundary**: H1 (supported at 0.1-nat resolution by $1.2 \cdot 10^9$ candidates), H2 (posed, Open Problem 7.4), the $o(1)$ terms and equidistribution inputs, and mechanisms outside $\mathcal{L}$ altogether — neither a planted divisor identity nor per-instance discovery — of which none is known.

The programme consequence is clean: with the covering monopoly a result, record progress is governed by Door 1 and Door 2 — and the honest forecast for each game is Table 6.5, unchanged by anything a mechanism was hoped to deliver.

Chapter 8

Conclusions, open problems, and the honest epitaph

We collect what the thesis established — proved, measured, and conjectured, marked apart — restate the economics in one paragraph, rank the open problems the work leaves behind, and record the methodological practices that kept a record-driven, heavily automated research programme honest.

8.1 What was established

Proved. $\mathfrak{g}$ is unbounded, with explicit partitions, by primordial obstruction plus Dirichlet (Theorem 2.6); any prime-offset cover is realisable by an actual Goldbach number (Theorem 2.12); the maximal order of $\mathfrak{g}$ is $\gg \log N \log_2 N \log_4 N / \log_3 N$ along explicit-partition integers (Theorem 2.15); shared divisors of two complements are supported below Q (Proposition 7.3); and, at draft rigor on the swept grammar, the Divisor-Paradigm Closure Theorem (Theorem 7.2): value-set ceilings, the cluster bound $k_{\max}$, isolation and orbit accounting, the ExpFamily purchase bound, and Compose emptiness — covering is the unique economical planted mechanism in $\mathcal{L}$.

Certified. The record board of Table 1.1, and behind it eleven new constructions (Table 4.1 and the two height records) each carrying unconditional exclusion witnesses, a certified partition, dual-stack verification, and manifests: the height records $\mathfrak{g} = 1\,157\,341$ (2 480 digits) and $\mathfrak{g} = 1\,134\,871$ (2 692 digits, with its 8 970 local gap certified, ratio 126.5); the threshold descent to N_{150} with $\mathfrak{g} = 104\,527$; the budget record $\mathfrak{g} = 119\,419$ under 10^{199} .

Measured. The cost model $E = |U| \cdot \text{boost} / \ln N$ to ± 0.1 nats over $1.2 \cdot 10^9$ candidates at five scales; the frontier curves $E(D)$, $E(Q)$, and the million-desert wall (10^{85} tests at 199 digits against $\sim 10^{6.5}$ existing targets); the engine optimizations ($1.8\times$, $3\times$, $430\times$); the compression ceilings (entropy-exact k -list cancellation; $H^{0.35}$ conditioning saturation; the Johnson radius as a practical wall at $1.028 A_J$); the assignment landscape (fractional coverage 100%; greedy at $z = 24.3$; 867 window-certified against exact solvers up to half the cover); and the adversarial-search ceiling per mechanism family (best corrected fitness 0.0919, $53\times$ short).

Conjectured, named. H1 (Hardy–Littlewood independence, supported at 0.1-nat resolution); H2 (Open Problem 7.4); the $o(1)$ terms and equidistribution inputs of Lemma 1; out-of-grammar

mechanisms; and the calibrated truths — $T(10^5) \approx 10^{53}$, $R(10^{200}) \approx 1.8$ million — that give the records their scale bar.

8.2 The three currencies, one paragraph

A large least Goldbach summand is bought with three currencies — modulus digits, search throughput, and certification compute — trading at measured rates: about $2.5\times$ search per 7 digits near 190, a digit per percent of cover quality, exponent $e^{|U|\cdot\text{boost}/\ln N}$ for everything. Complete covers were the gold standard and are obsolete; partial covers plus brute progression scanning, checkpointed in git and certified twice by independent code, are the entire present frontier. The truth for $T(10^5)$ sits a factor ~ 3 below the record in logarithm, and the combined barrier analysis says only 10–20% of that log-gap will fall to compute and cover quality, with everything below ~ 120 digits protected by two independent entropy walls and a closure theorem. The records are what they are: *upper bounds, manufactured, falling toward a floor they will never touch* — a sentence that belongs in every future README of this genre.

8.3 Open problems, ranked

Ranked by expected impact per unit effort, with the chapter that motivates each.

1. **The rounding-gap programme** (§7.8). Determine the achievable residual fraction for one-class-per-modulus covers of $\pi(Q) \approx 10^4$ primes under budget $\theta(457)$ — importing the Erdős–Rankin/FGKMT layered covering constructively at finite scale. Every absolute percent is $\approx 15\times$ compute; half the greedy-to-fractional gap is $\approx 22\,000\times$. Includes: certify the quenched threshold (1RSB/survey propagation with complexity bookkeeping); lift-based bounds (Sherali–Adams on the small-moduli block) where the plain LP is provably vacuous; and the banded-ensemble construction with an exact matching endgame.
2. **H2: sub-modexp compositeness detection** (Open Problem 7.4). Either direction is a result: a detector re-prices every prime-hunting search on earth; a lower bound in any realistic model finishes Lemma 3.
3. **The two-sided construction** (§5.6). Intersect cover congruences with a certifiable-by-construction complement family (Proth/Pocklington). Uncaps the height game — the only game with no mathematical wall — from ECPP practice to Proth-search scale. Key unknown: Proth-prime density inside CRT-constrained progressions.
4. **CRT list-recovery beyond the Johnson radius** (§6.4–6.5), for structured, solution-abundant instances. The measured wall at $1.028 A_J$ says practice needs *new decoding*, not better reduction.
5. **Modular subset-sum with choices at density ≈ 1** when 2^{82} solutions exist (§6.4). Progress on either 4 or 5 collapses the sub-100 threshold verdict.
6. **Aggregate compositeness certificates**. Verification is linear in the witness count; is there a sublinear batch certificate that all k complements are composite (resultant/product-tree/gcd object verifiable faster than k strong tests)? Even $5\times$ changes height-game economics; an impossibility result would be a certificate-complexity theorem. Untouched.

7. **Matching lower bounds.** There is no theorem of the form “every even $N < B$ has $\mathbf{g}(N) \leq f(B)$ ” beyond trivialities, even conditionally. A GRH-conditional explicit bound, or a large-sieve bound on how many residue classes a small N can occupy, would turn Chapter 6’s walls from forecast into mathematics.
8. **Complexity placement.** Is deciding $\exists \text{even } N < B : \mathbf{g}(N) > Q$ (primality oracle given) NP-hard? The reductions of Chapter 6 are evidence about *attacks*, not the problem; either answer locates the game on the map.
9. **Neighbouring games, all virgin.** $S(q)$ exactly (the smallest desert with prescribed least summand — posed in 1989, no constructive records exist); double deserts (force the two smallest summands large); Lemoine deserts (odd $N = p + 2q$, force $\min p$); ternary Goldbach deserts. The pipeline sets first-ever certified marks in each essentially for free.
10. **Science from the exhaust.** The campaigns banked $\sim 10^9$ PRP tests of structured sparse sets at heights 10^{150} – 10^{2700} with full logs — the largest such sample ever produced. Testing Hardy–Littlewood k -tuple corrections against fail-position statistics is a free empirical paper and a recalibration of the p_1 model everything rests on.
11. **Formal verification.** The standalone verifier is ~ 200 lines of elementary arithmetic; a Lean/Coq proof of “exit 0 $\Rightarrow \mathbf{g}(N) = q$ ” would make these the first machine-*proof*-carrying records in the genre and close the last trust gap.
12. **Correlation-aware covers.** Deliberately designing the residual set for inadmissibility (minimal singular series) provably raises the all-composite probability at fixed $|U|$; P5 bounds the *accidental* version at 0.1 nats, so expect a modest constant — but nobody has optimised for it.

For the engineering track, the costed roadmap (“Operation Staircase”) stands ready: rungs at $Q = 140\text{k}, 150\text{k}, 165\text{k}, 175\text{k}, 200\text{k}$ under 10^{200} , each banked and dual-certified, with the final $e^{5.5}$ wall closed by any one of volunteer computing, an ASIC rack, or two nats of Problem 1 — and a committed variant-ledger so parallel workers scan disjoint space. The 2.5-million height rung awaits only its (priced, one-time) 3 300-digit certification.

8.4 Methodological legacy

Five practices earned permanent status, and are offered to any record-driven computational programme:

1. **Verification-first.** Witnesses re-derived from published data by standalone code, dual implementations for every certificate, counts asserted against independent totals, manifests over everything. The two false-PASS incidents (§4.5) were caught by redundancy, not luck; no five-leg record was ever overturned.
2. **Adversarial audits, of ourselves first.** Planted corruptions against the verifier; a hostile review of a contributed blueprint before adoption; and the falsification engine’s referee broken by its own adversary within ten candidates — the fix strengthening the theorem it guarded. Never let a candidate report its own fitness; never trust a pass without hands.

3. **Honest negatives, with statistics.** Hitless campaigns reported with survival probabilities and per-slice model agreement; one retraction (§6.4) documented with its root cause — two bounds optimised at different parameters — as a lesson, not buried.
4. **Compute that survives its substrate.** Slices as transactions, checkpoints in version control committed before results, harvest-first workers: ~ 25 container losses across the programme, zero coverage lost.
5. **Models before campaigns.** No search was launched without a validated density forecast, a $3\times$ budget, and written stopping rules; every hit and every miss then updated the calibration. The model never needed post-hoc adjustment — which is the sentence that makes all the others cheap.

8.5 Coda

The moral is a trade-off, not a paradox — and by now it is a measured one. Nature keeps the least Goldbach summand microscopically small, and believes (1.2) it never grows much faster than $(\log N)^2 \log \log N$. Against that backdrop, deserts can be manufactured: with fully static certificates at a thousand digits, or with controlled search at a hundred and fifty, or past every certified prime gap at twenty-five hundred — each purchase itemised in a currency whose exchange rates this thesis measured, and whose counterfeits — shortcuts through lattices, conditioning, or algebra — it priced honestly at infinity. What remains between the records and the truth is now a short, explicit list of mathematical problems rather than a fog. That list is the thesis’s real deliverable: cute gaps, and precisely posed ways to make them cuter.

Appendix A

Artifact map and verification runbook

Everything in this thesis is re-checkable from the artifact repository (the `goldbach/` and `slop/goldbach/` trees). This appendix maps claims to artifacts and gives the one-command verification for each record. Environment as tested: Python 3.11 with `gmpy2`, `numpy`, `sympy` (and `python-flint` for the lattice study), PARI/GP 2.15.4 [22]; the `slop` verifier runs on the Python standard library alone.

A.1 Records and their verifiers

Each record directory contains `record.json` (construction data: cover pairs, N_0 , M , k , the decimal N), the evidence file (per-offset witnesses), the ECPP certificate(s) (`*.gp`), the validation log, and a SHA-256 manifest.

In the table, `G-records/` abbreviates `goldbach/records/` and `S-records/` abbreviates `slop/goldbach/records/`; the GPU/goldbach-pipeline records verify with `goldbach/verify_record.py` (the mega-desert with `goldbach/verify_megagap.py`), the CPU/slop-pipeline records with the stdlib-only `slop/goldbach/records/verify_record.py`.

record	directory	stack
<code>g = 104 527, 150d (T)</code>	<code>G-records/threshold_150digit_g104527</code>	G
<code>g = 100 297, 195d</code>	<code>G-records/threshold_195digit_g100297</code>	G
<code>g = 107 719, 197d</code>	<code>G-records/dual_197digit_g107719</code>	G
<code>g = 112 249, 200d</code>	<code>G-records/budget_1e200_200digit_g112249</code>	G
<code>g = 1 134 871, 2692d</code>	<code>G-records/megagap_2692digit_g1134871</code>	G (megagap)
<code>g = 1 157 341, 2480d (H)</code>	<code>S-records/g1</code>	S
<code>g = 110 917, 199d</code>	<code>S-records/g23</code>	S
<code>g = 102 337, 193d</code>	<code>S-records/t100k (round 1)</code>	S
<code>g = 109 357, 186d</code>	<code>S-records/t100k (round 2)</code>	S
<code>g = 101 149, 179d</code>	<code>S-records/t100k (round 3)</code>	S
<code>g = 119 419, 199d (R)</code>	<code>S-records/r200</code>	S

Typical invocations:

```
python3 goldbach/verify_record.py \  
    goldbach/records/threshold_150digit_g104527/record.json /tmp/ev.json  
python3 goldbach/verify_megagap.py          # full megagap re-check + manifest  
python3 slop/goldbach/records/verify_record.py \  
    slop/goldbach/records/g1/record_g1.json
```

The verifiers reconstruct N by CRT, re-derive every witness below $\mathbf{g}$ (asserting the scanned count against an independent $\pi(\mathbf{g})$), re-verify the ECPP chain from scratch including the binding of its top integer to $N - \mathbf{g}$, and check the record inequalities; exit code 0 means all legs pass. ECPP certificates additionally validate under PARI's `primecertisvalid`, and the independent projective checker is `goldbach/ecpp_check.py` (`slop/.../g23/ecpp_verify.py` for the CPU-session records, which also carry APR-CL logs and `crosscheck_*.gp` full-scan scripts).

A.2 Search, cover, and analysis tools

Paths below are relative to `goldbach/`.

tool	role (chapter)
<code>cover.py</code> , <code>anneal.py</code>	greedy weighted covers; SA refinement (4)
<code>gen_variants.py</code>	residue-swap variant catalogs (4)
<code>search.py</code>	sieved progression scan; <code>--sort-tests</code> , <code>--skip-frac</code> (4)
<code>gpu/engine150.py</code> , <code>gpu/fleet.py</code>	CUDA engine and fleet driver (4)
<code>verify_record.py</code> , <code>package_record.py</code>	witness generation; record packaging (4)
<code>joint.py</code>	joint representative/coverage optimizer, k -list meet-in-the-middle (6)
<code>exp1m.py</code>	million-desert frontier and wall (3)
<code>entropy_frontier.py</code> , <code>realizable_frontier.py</code> , <code>frontier_table.py</code>	set-valued conditioning study (6)
<code>verify_barrier.py</code>	existence-vs-decoding coupling table (6)
<code>phd/toy_crt/decoder.py</code>	LLL vs Johnson radius study (6)
<code>phd/wp8/lp_bound.py</code> , <code>phd/wp8/pack_probe.py</code>	LP vacuousness; packing probe (7)
<code>phd/wp9/engine/</code>	closure-theorem engine: <code>tier0.py</code> , <code>exhaustive.py</code> , <code>scorer.py</code> , harness docs (7)
<code>phd/wp9/door1/</code>	rounding-gap instance, CP-SAT LNS, BP/SP, verdicts (7)

Constructions referenced in Chapter 6 are committed under `goldbach/constructions/`: `m211_x196.json`, `m219_x197.json`, `m221_x197.json`, `wall_1M_199d.json`, and `blueprint_1M_1250d.json`, each re-verified end to end by its generator.

A.3 Research notes underlying each chapter

The narrative documents merged from the research sessions, kept as primary sources (paths again relative to `goldbach/`): `RESULTS.md` (record announcements and statistics), `SOK_GOLDBACH_GAPS.md` (systematization, walls), `FRONTIER.md` (set-valued conditioning and sub-100 wall), `IDEAS.md` and `IDEAS100.md` (the idea ledgers with kill-status), `PROPOSAL_NEXT_CAMPAGNS.md` (the stair-case), `phd/RESEARCH_LOG.md` (the dated log of campaigns, audits, and merges), the memo `phd/wp3_small_representative_memo.md` (subset-sum barrier, including the v1 retraction), `phd/wp7_probes.md` (the five probes), `phd/wp8_beyond_covering.md` (doors; rung ladder; LP result), `phd/wp9/wp9_closure_theorem.md` (the theorem memo, §1–14), `phd/wp9/h2_open_problem.md` (H2, posting-ready), and `phd/wp9/door1/` (landscape diagnostics and gate verdicts). Campaign drivers and their committed checkpoint logs live under `phd/campaign*/` and, for the CPU session, `slop/goldbach/records/work/`.

A.4 Independence disclosure

Two verification stacks were developed in separate sessions with no shared code: the `goldbach` stack (`verify_record.py` + `ecpp_check.py` + PARI) and the `slop` stack (stdlib-only `verify_record.py` + `ecpp_verify.py` + APR-CL + PARI full scan). Every headline record has passed both. The adversarial audit transcript (planted corruptions, all detected) and the two false-PASS incidents with their root causes are recorded in `goldbach/SOK_GOLDBACH_GAPS.md §4` and `goldbach/phd/RESEARCH_LOG.md`.

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